Course Announcements

Due Sunday (11:59 PM):

- A5
- Q6
- Project Checkpoint #1: Data
- Weekly Project Survey (*optional*)

Notes:

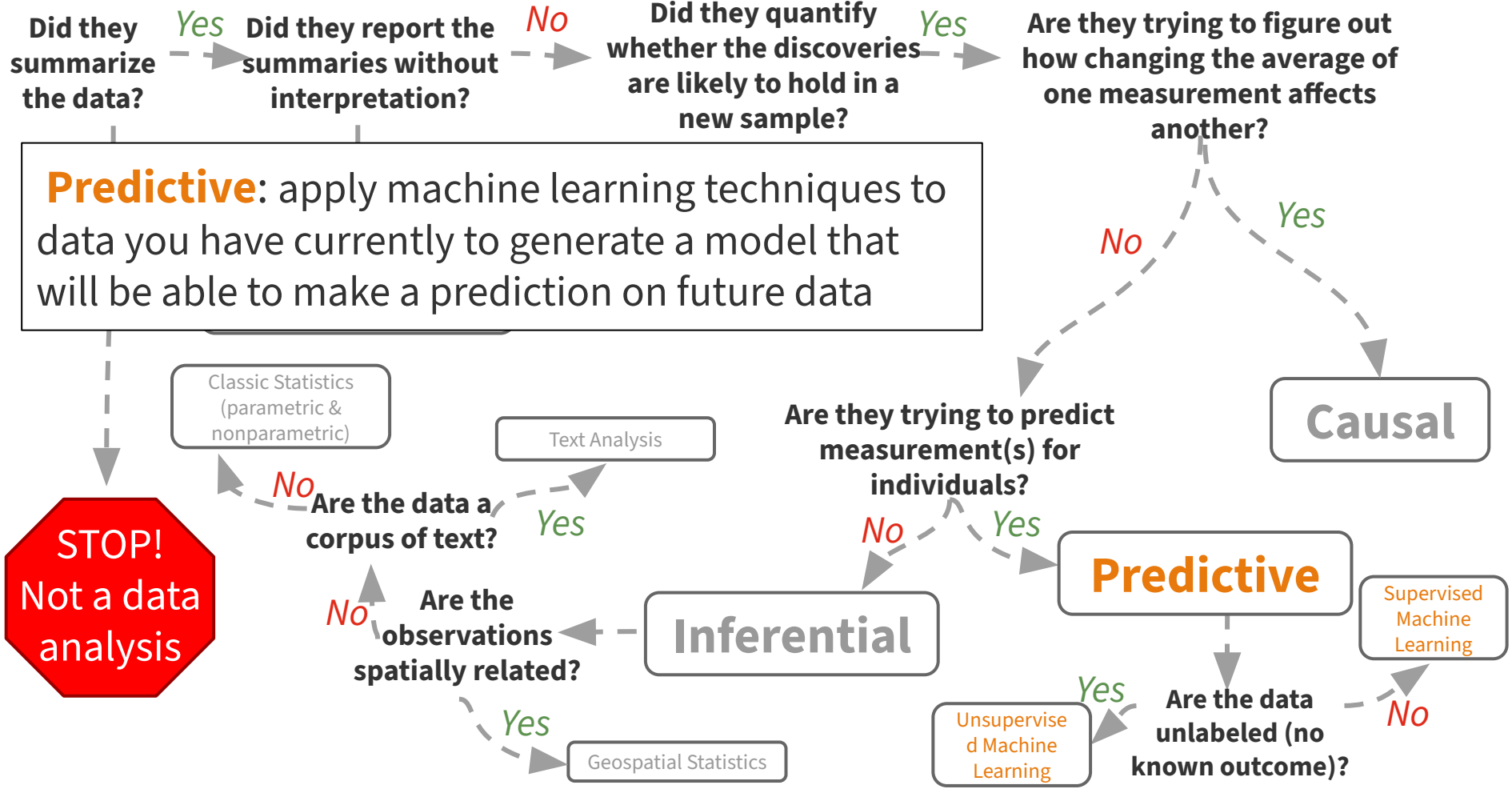
- Project Proposal Feedback + Score now on GitHub repo (as Issue)
  - Lost proposal points? Earn them back in your Data checkpoint (optional: add a “Response to Proposal Feedback” section)
- A6 (machine learning) releases Fri and uses today’s material

Machine Learning

Shannon E. Ellis, Ph.D
UC San Diego

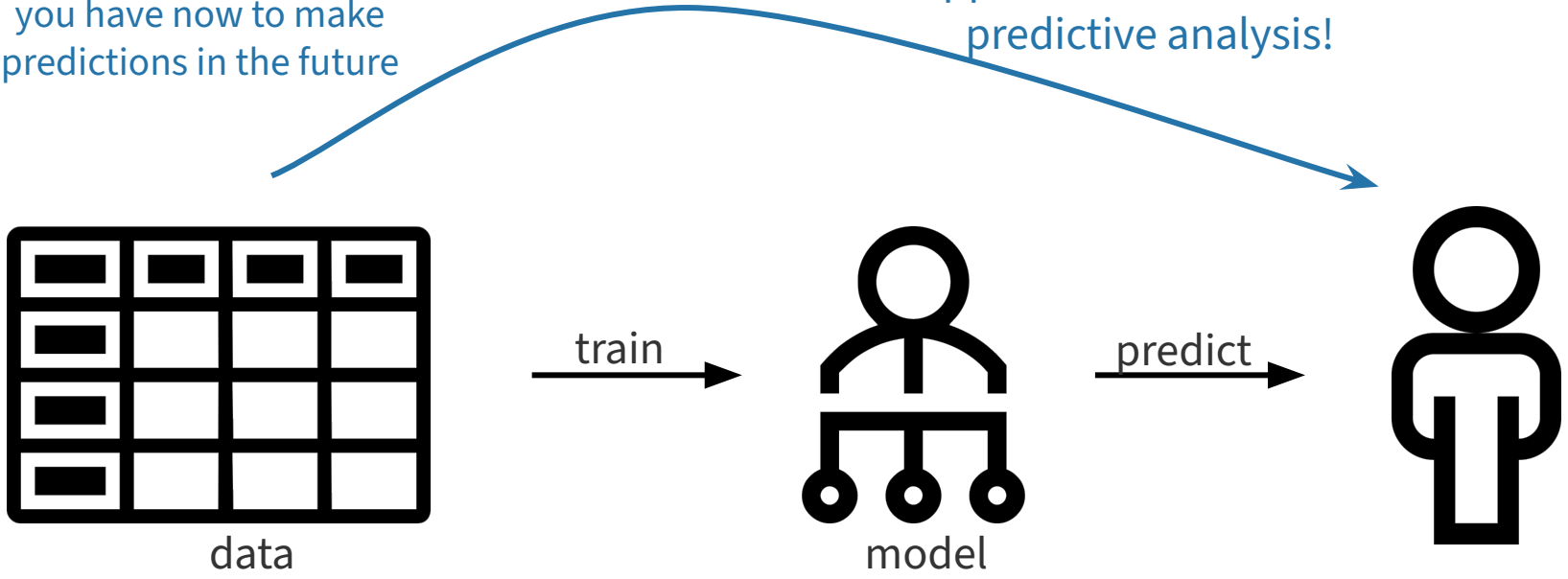
Department of Cognitive Science
sellis@ucsd.edu




Did they summarize the data?
Yes
Did they report the summaries without interpretation?
No
Did they quantify whether the discoveries are likely to hold in a new sample?
Yes
Are they trying to figure out how changing the average of one measurement affects another?
Predictive: apply machine learning techniques to data you have currently to generate a model that will be able to make a prediction on future data
No
Yes
Causal
Are they trying to predict measurement(s) for individuals?
No
Yes
Predictive
Supervised Machine Learning
Are the data unlabeled (no known outcome)?
Yes
Unsupervised Machine Learning
No
Inferential
Are the observations spatially related?
No
Yes
Geospatial Statistics
Are the data a corpus of text?
No
Yes
Text Analysis
Classic Statistics (parametric & nonparametric)
STOP! Not a data analysis

predictive analysis uses data you have now to make predictions in the future

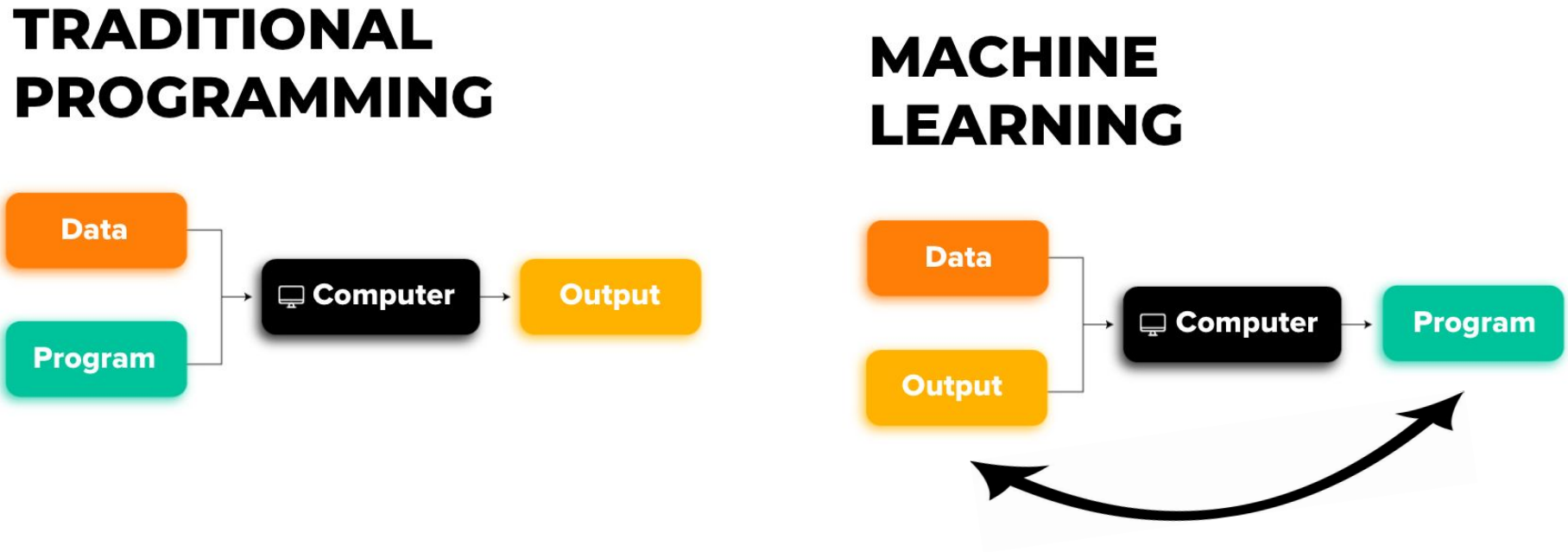
machine learning approaches are used for predictive analysis!



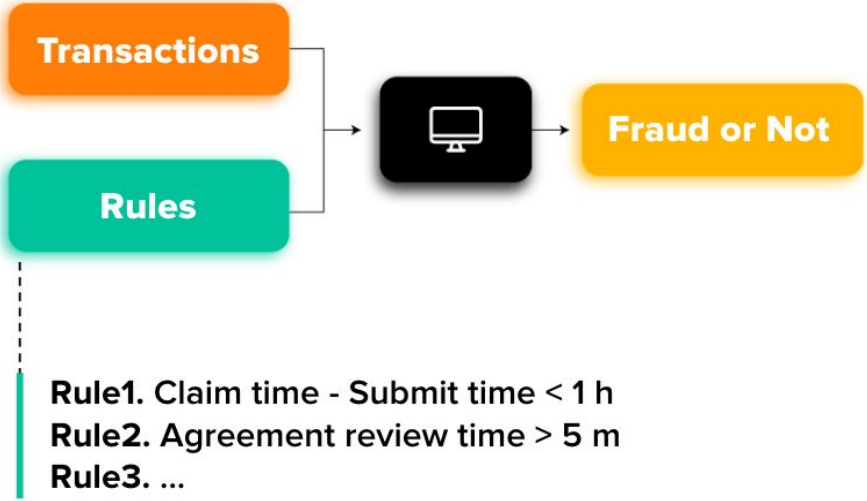
What is machine learning?

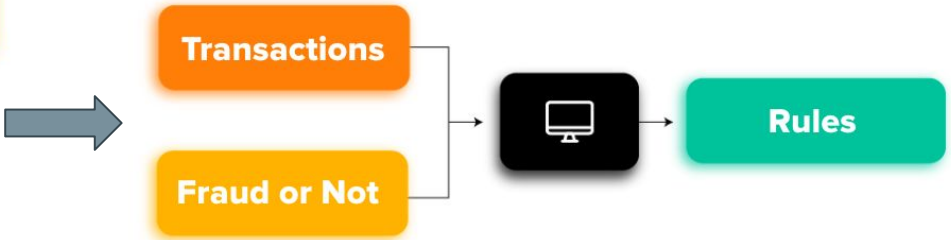
> “*Machine learning is the science of getting computers to act without being explicitly programmed*”
>
> - Andrew Ng (Stanford/Coursera/Google Brain)

A Shift in Programming Paradigm



https://www.avenga.com/magazine/machine-learning-programming/





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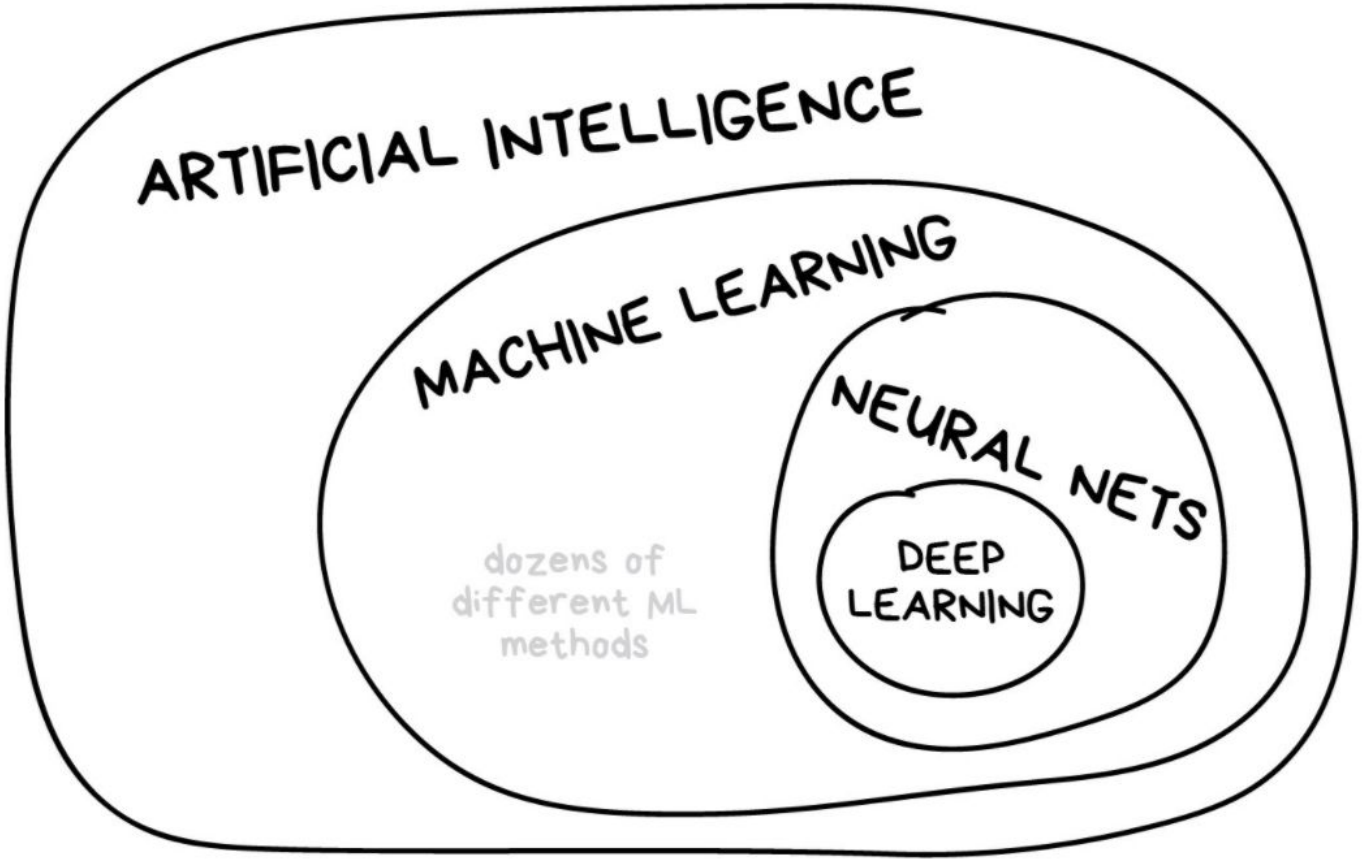


Image source: https://vas3k.com/blog/machine_learning/

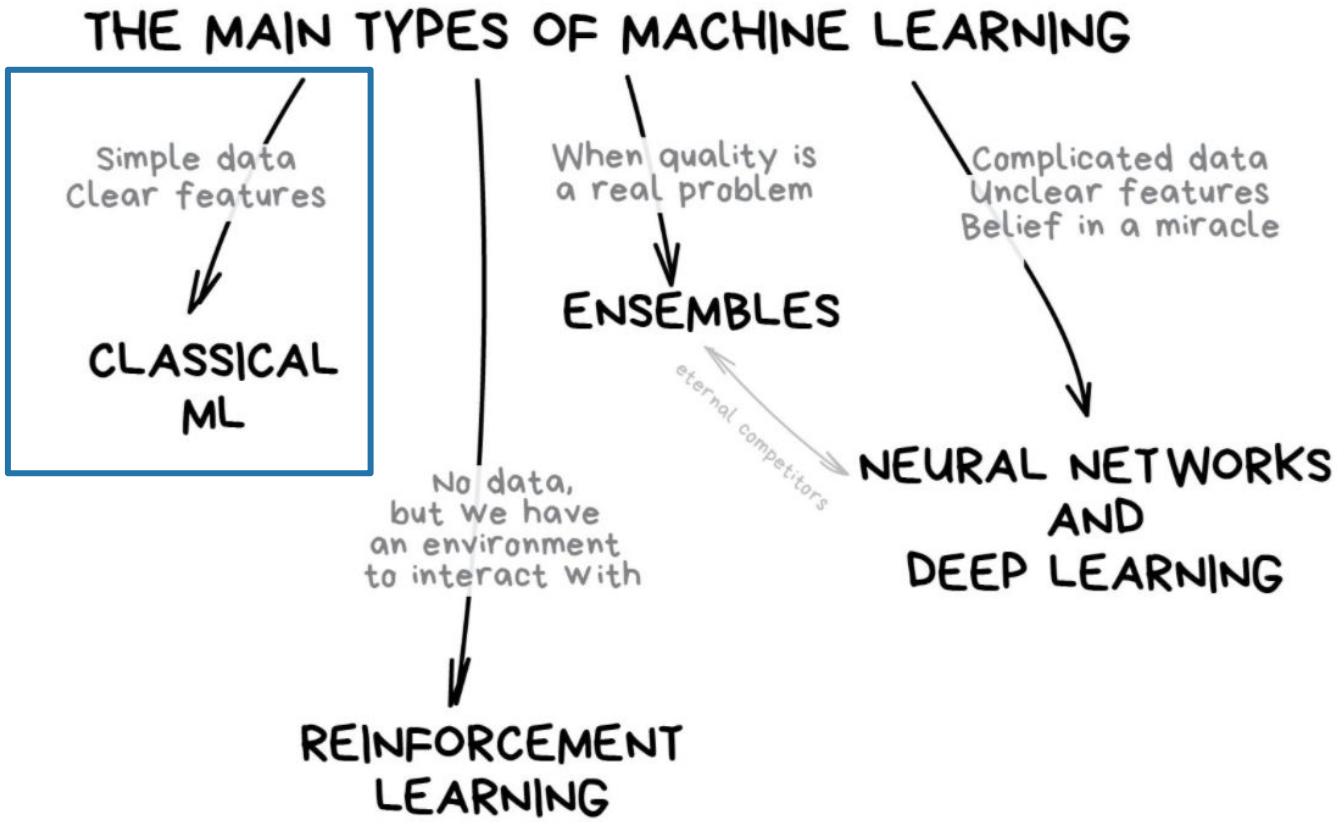
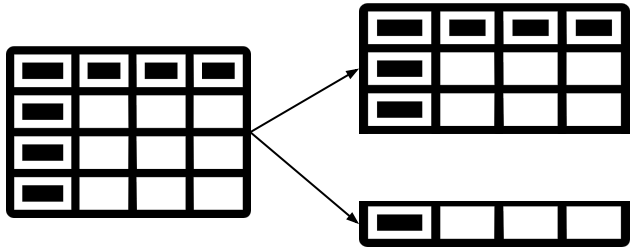
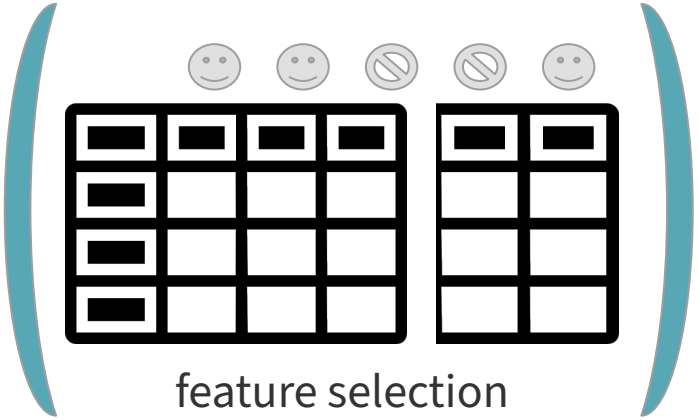


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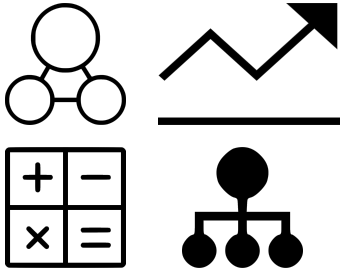
Basic Steps to Prediction



data partitioning



feature selection



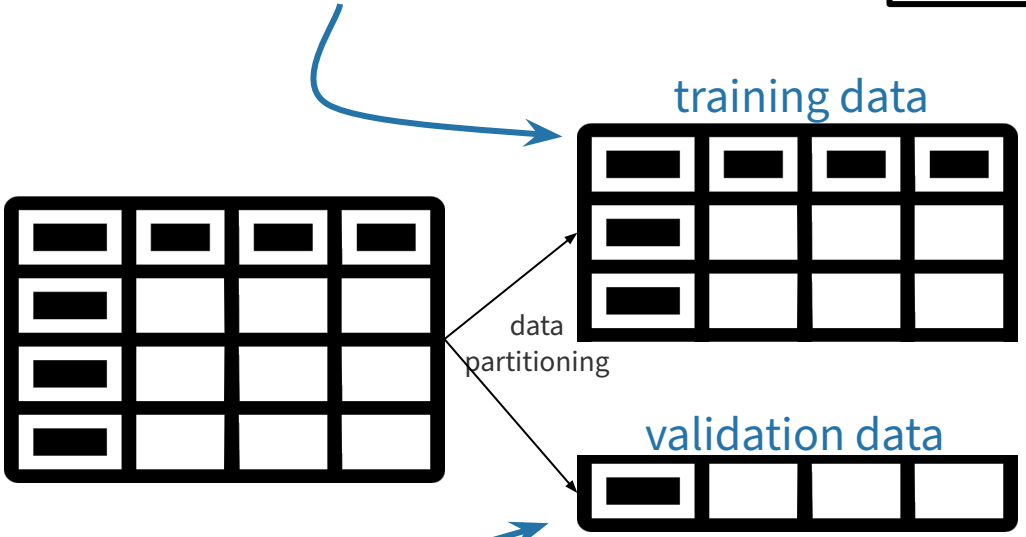
model selection



model assessment

Data partitioning: Evaluating how you will do when you have large amounts of data

the data used to build your predictive model

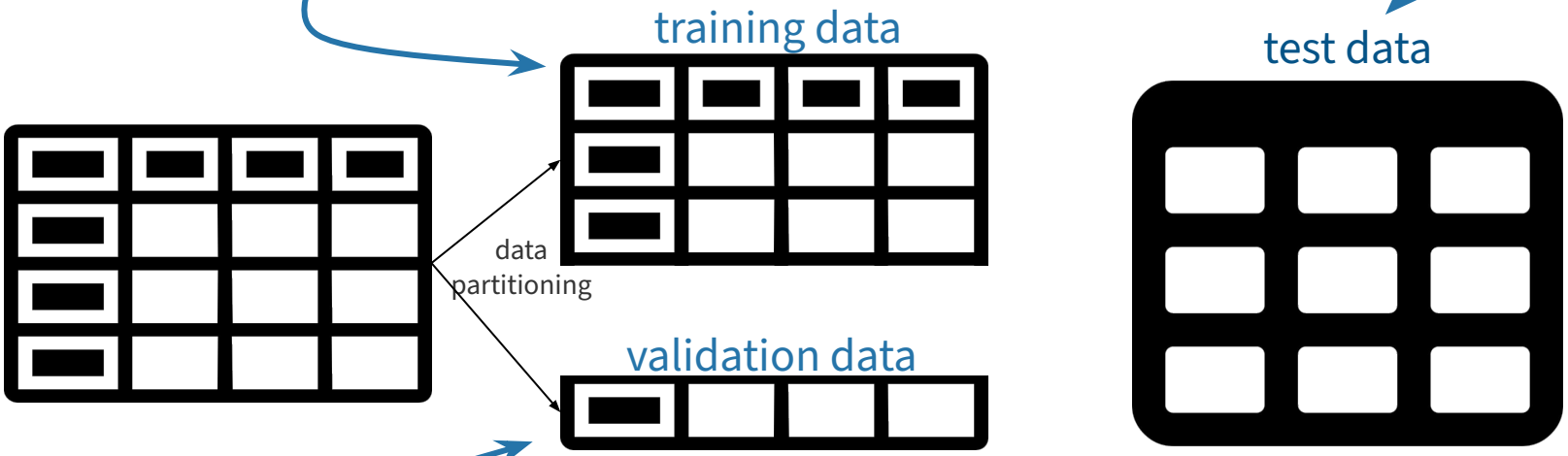


Data from original dataset that was held out and not used in training the model ; helpful in fine-tuning prediction accuracy

Data partitioning:
Finding the best model through validation

the data used to build your predictive model

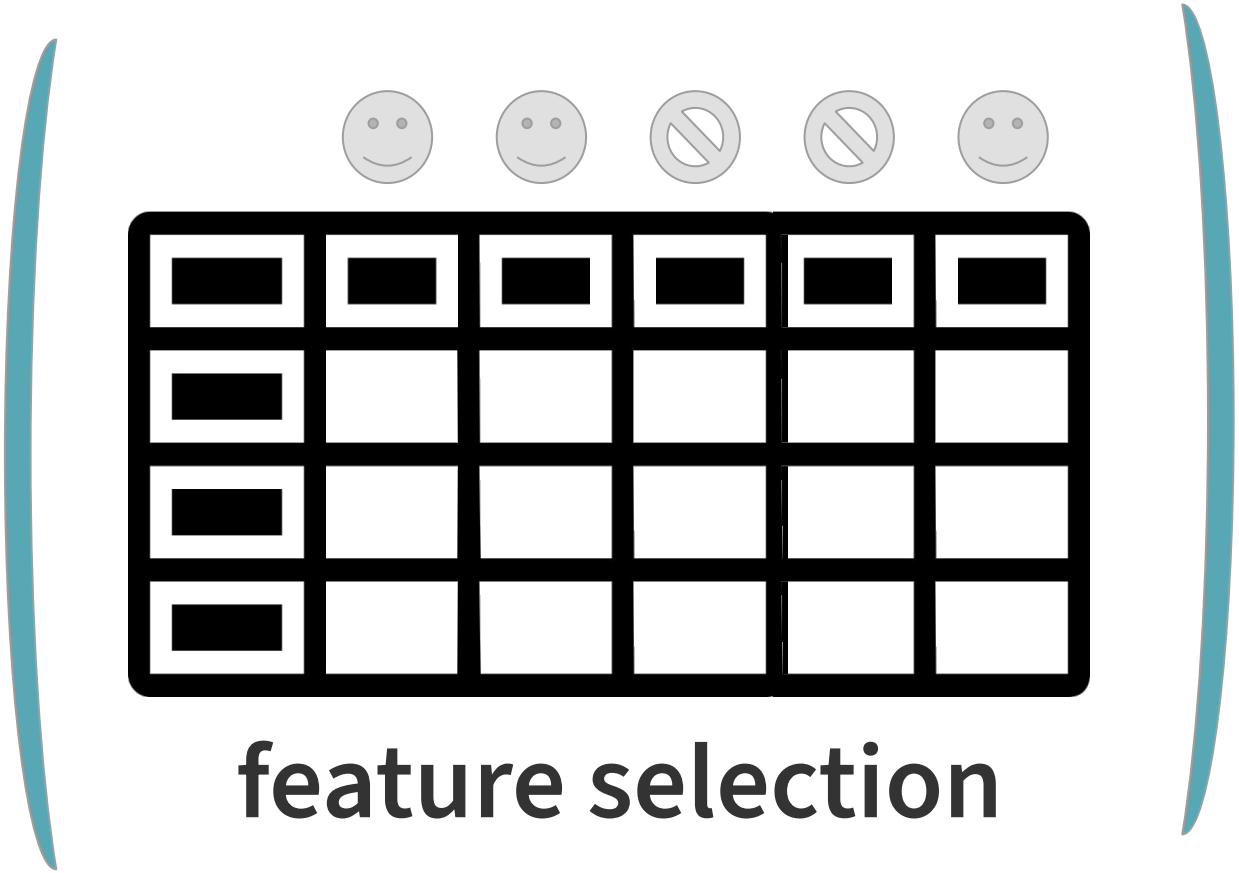
new and independent data set used to assess if prediction model is generalizable



Fine-tune the model to increase prediction accuracy (Hyper-parameter tuning)

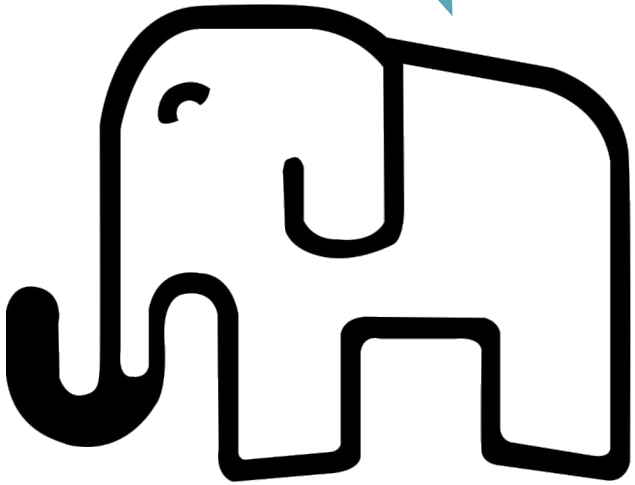
Parameters vs. hyperparameters

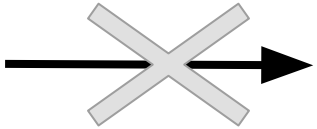
- Parameters are learned from the training data (e.g., the slope and intercept of a regression)
- Hyperparameters are settings you choose before training (e.g., k, the number of neighbors in KNN)
- Choose hyperparameters with validation data or cross-validation
- Never use the test set to choose them: save it for the final check



feature selection

elephant height data are likely not predictive of US elections



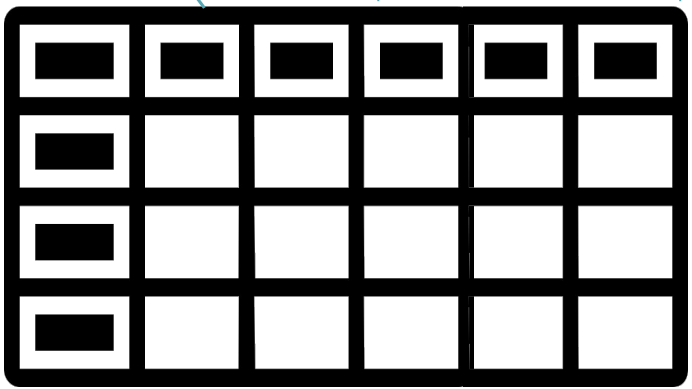




previous elections

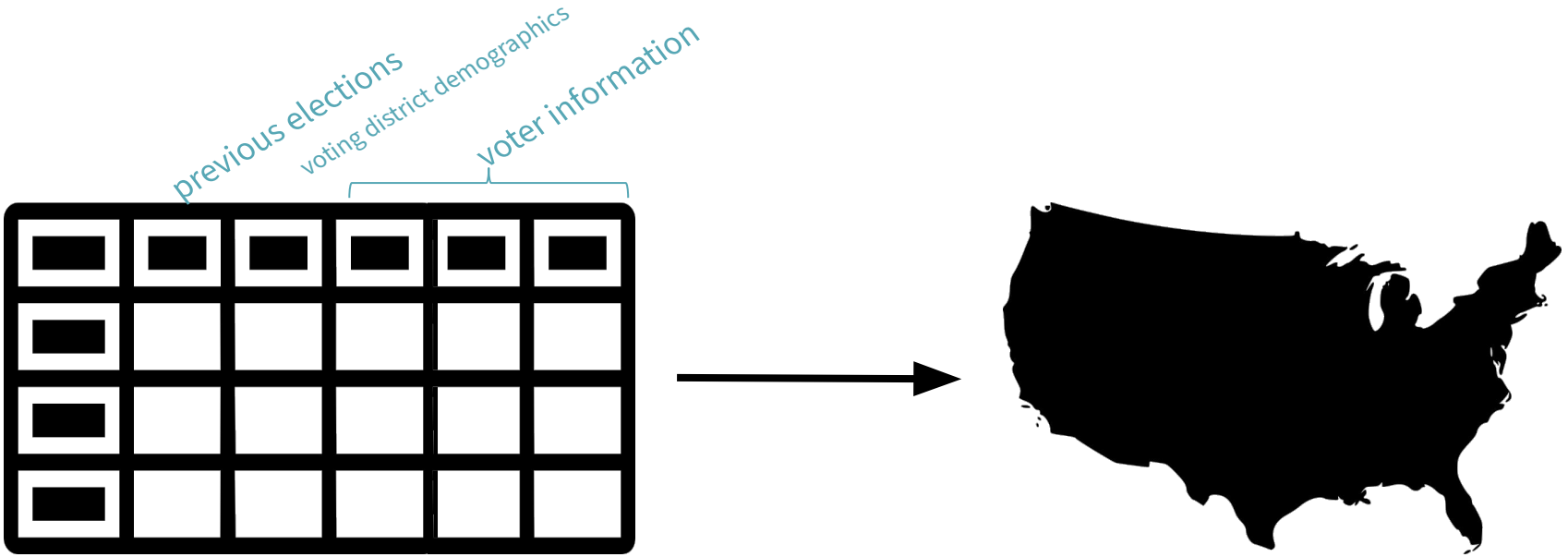
voting district demographics

voter information

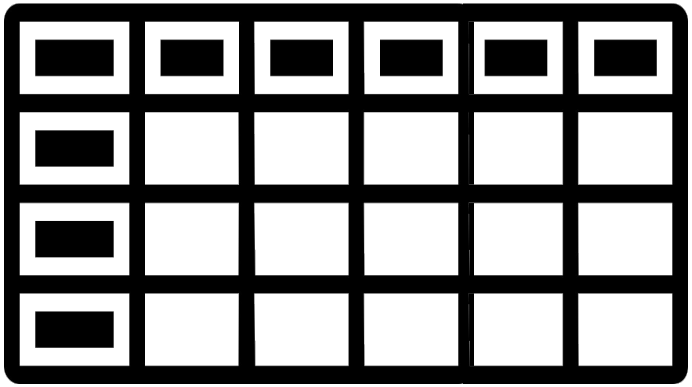




these data are likely predictive of US election outcomes



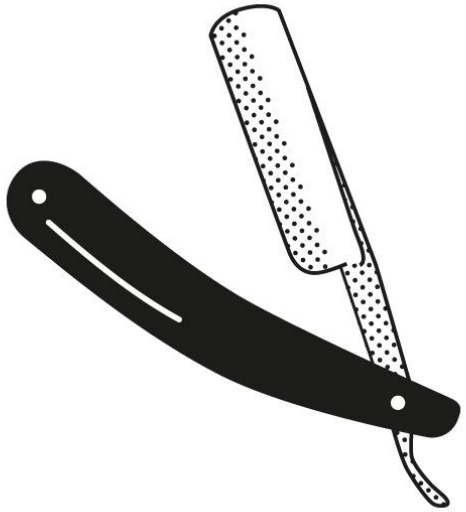
feature selection determines which variables are most predictive and includes them in the model



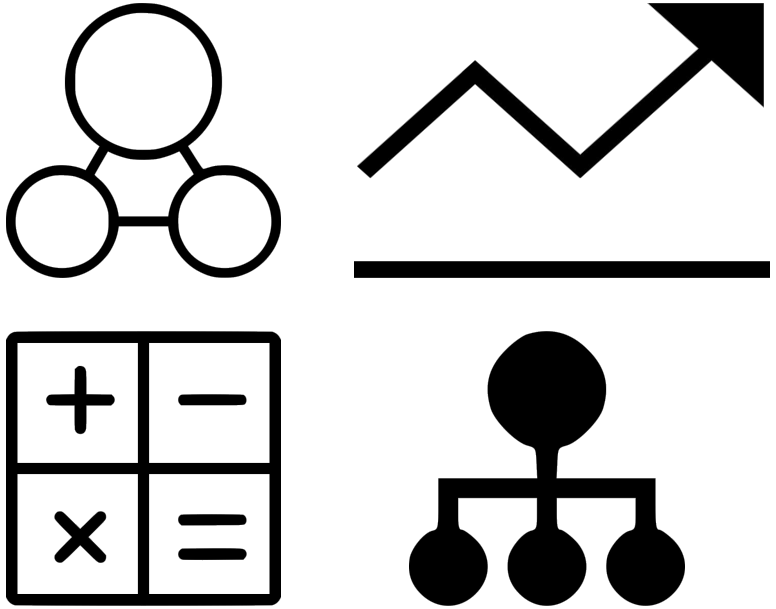


variables that can be used for accurate prediction exploit the relationship between the variables but do NOT mean that one causes the other

Note: Occam’s Razor



- When faced with two explanations for the *same evidence*, we prefer the explanation that makes the fewest assumptions
- Given *equal performance*, choose the simpler model



model selection

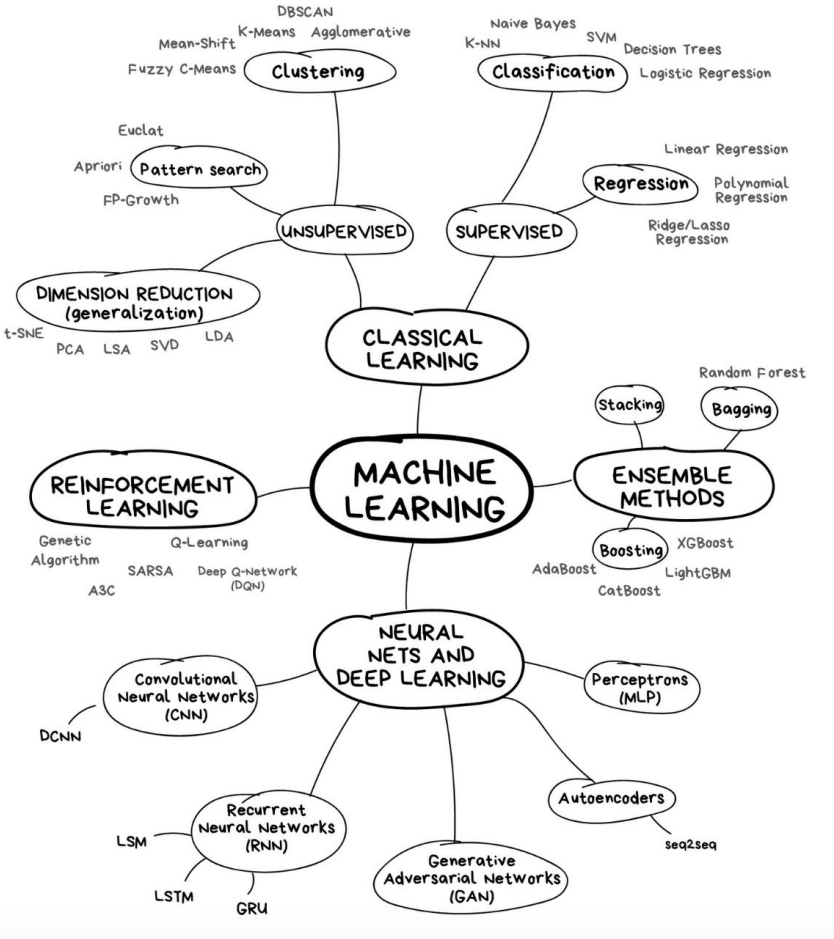
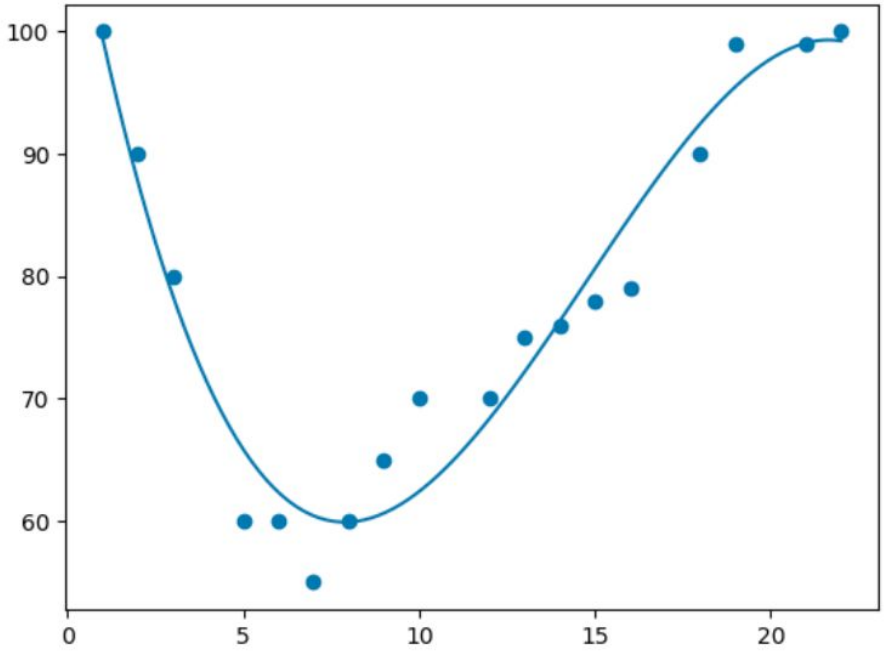
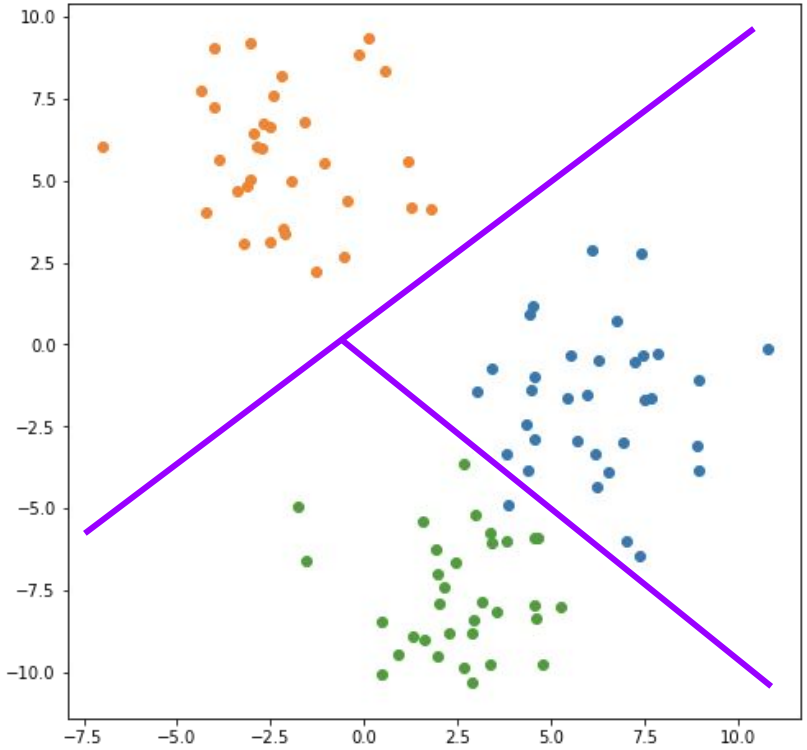


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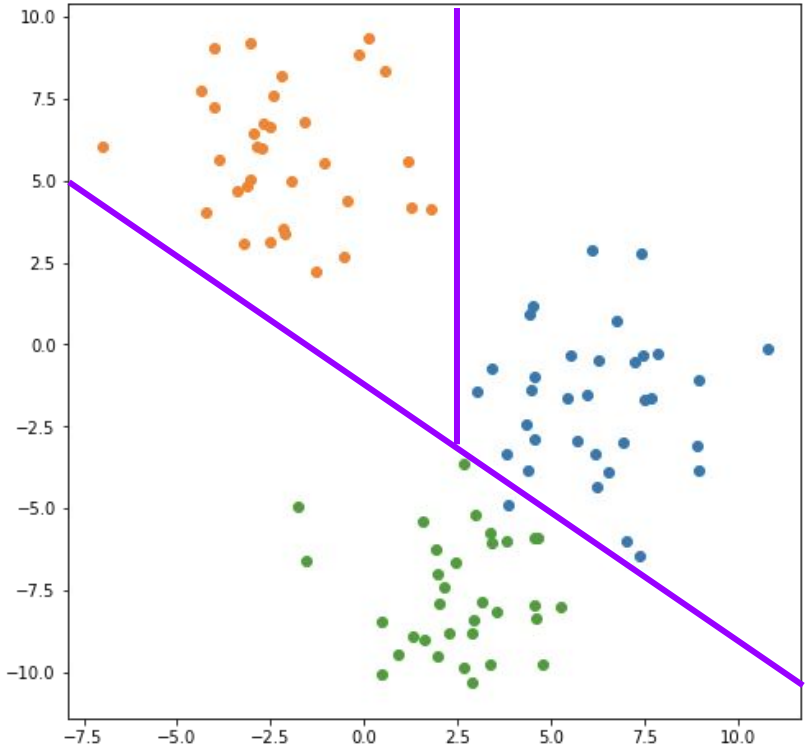
Selecting the “Right” Model



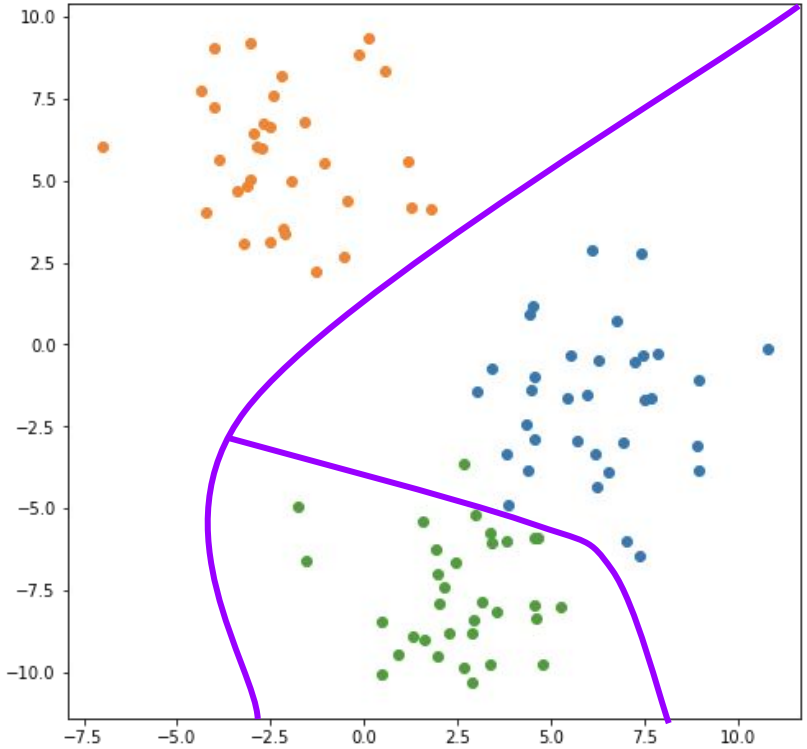
Selecting the “Right” Model



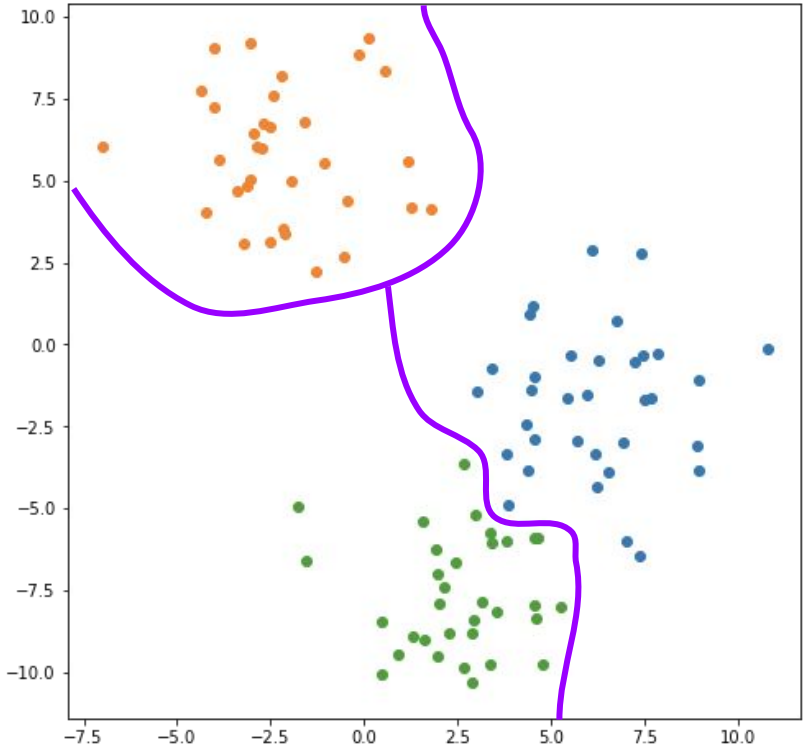
Selecting the “Right” Model



Selecting the “Right” Model



Selecting the “Right” Model



Cross-Validation

In reality, our eyeball meter won't always (and sometimes can't) cut it

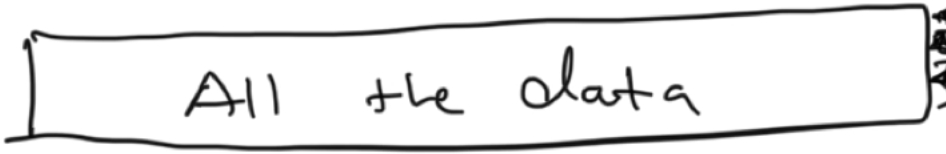
Real data are messy and live in dimensions we cannot even comprehend (let alone visualize)

Cross-validation offers a systematic way of assessing various models and determining which ones meet our requirements the closest

- Validation Set
- Leave-one-out
- K-Fold

Bigger Datasets. Simpler Models.

1



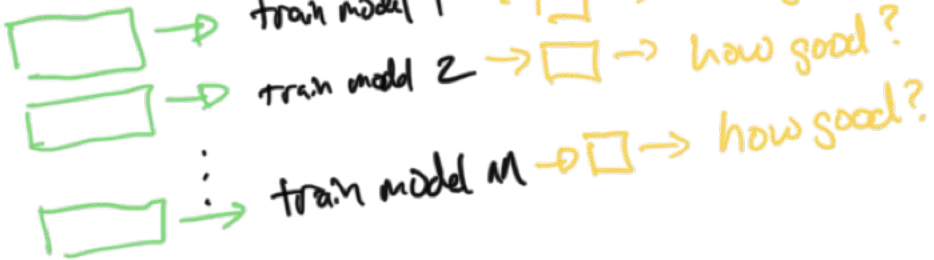
randomly shuffle rows of dataset

2



Split into training, validation and test sets

3



- if you have M models to try do this once for each

4



Pick best model from validation set performance

train it on T+V then test it on Test set

Model, feature, and hyperparameter selection

Needs: Don’t leak information from training/selection process into the test set

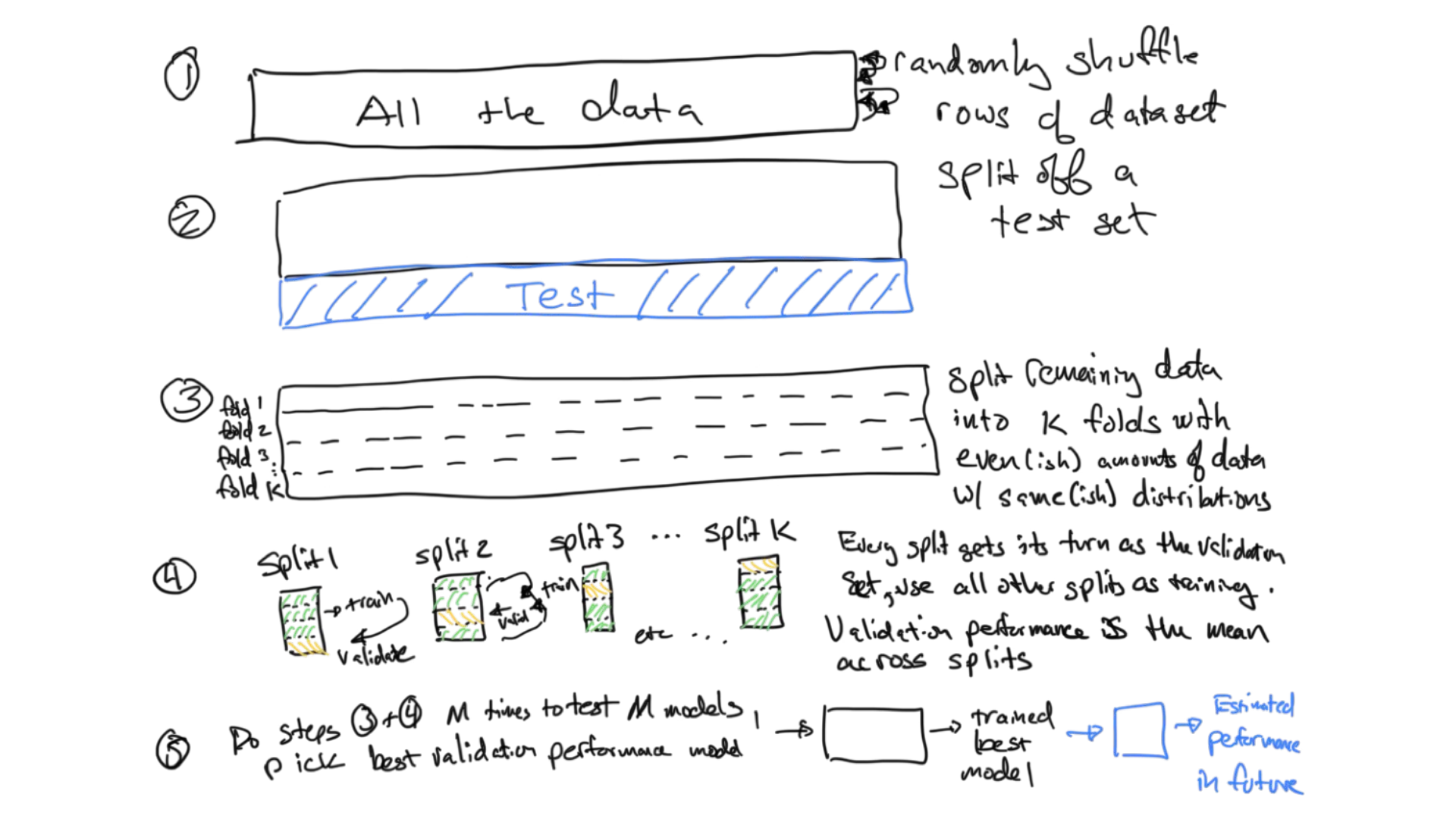
Trade-offs: Usually not enough data to have completely separate train, validation, test sets - where to prioritize?

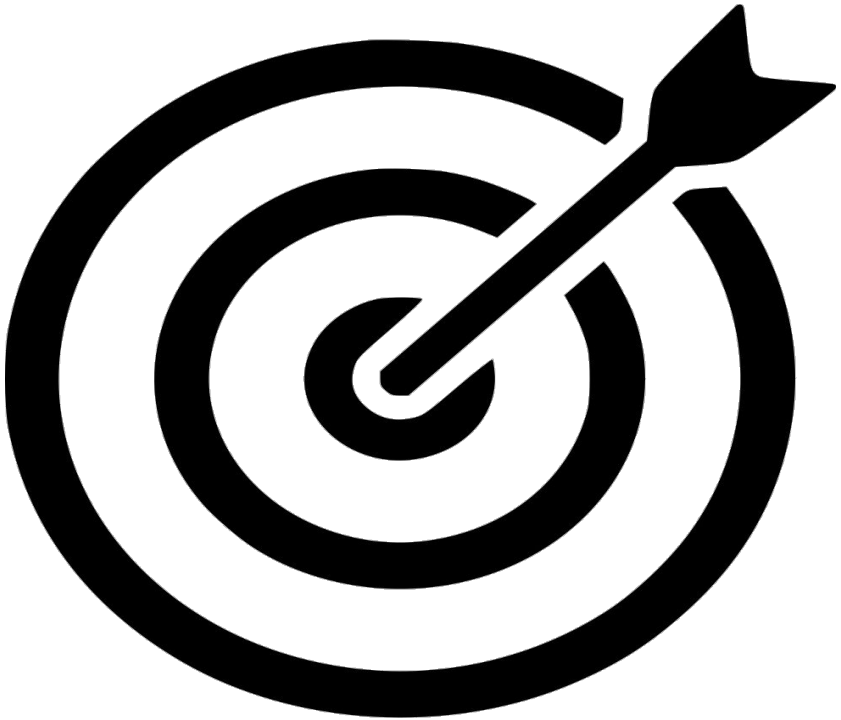
- Low training data == bad fit
- Low validation data == bad selection of model/features/hyperparameters
- Low test data == poor estimate of generalization performance

Fit preprocessing on the training data only

- Scaling, PCA and feature selection all learn something from the data (means, standard deviations, components)
- Fit them on the training set, then apply that same transformation to the test set
- Fitting on all the data first leaks test-set information into the model, so test scores look better than they will on new data
- A scikit-learn Pipeline (scaler → PCA → model) fits every step on the training data for you; you'll build one in A6 Part 4

Smaller Datasets. Complex Models.


1
All the data
randomly shuffle rows of dataset
split off a test set
2
Test
3
fold 1
fold 2
fold 3
fold K
split remaining data into K folds with even(ish) amounts of data w/ same(ish) distributions
4
Split 1
Split 2
Split 3
... Split K
train
validate
train
valid
etc ...
Every split gets its turn as the validation set, use all other splits as training.
Validation performance is the mean across splits
5
Do steps 3+4 M times to test M models, pick best validation performance model
trained best model
Estimated performance in future



model assessment

Many Metrics!

Assessing machine learning models involves gauging their performance on novel data

- Novel data can come from testing or validation sets
  - Validation → Fine-tuning parameters
  - Test → Final assessment before deployment
- Performance assessments depend on the specific problem/question
- Different metrics measure different aspects of models
  - e.g. MSE/RMSE, MAE (continuous); accuracy, precision, recall, specificity (categorical)

https://forms.gle/AeRuy7WrP6JZWR5t5

Q1: Chance performance

1 in 1,000 people have a disease. A “model” that predicts healthy for everyone has an accuracy of:

A) 0.1%

B) 50%

C) 99%

D) 99.9%



Accuracy can mislead

If classes are imbalanced, what would “chance performance” be?

Classic example: Cancer detection

- 1/1000 actually have cancer
- Your prediction algorithm misdiagnoses 1% of people
- Algorithm will tell 10 people who are healthy they are sick (FP) for every 1 person who is actually sick (TP)
- A model that predicts “healthy” for everyone is 99.9% accurate and never finds a single case
- Only 1 in 11 positive results is real: high accuracy hides this

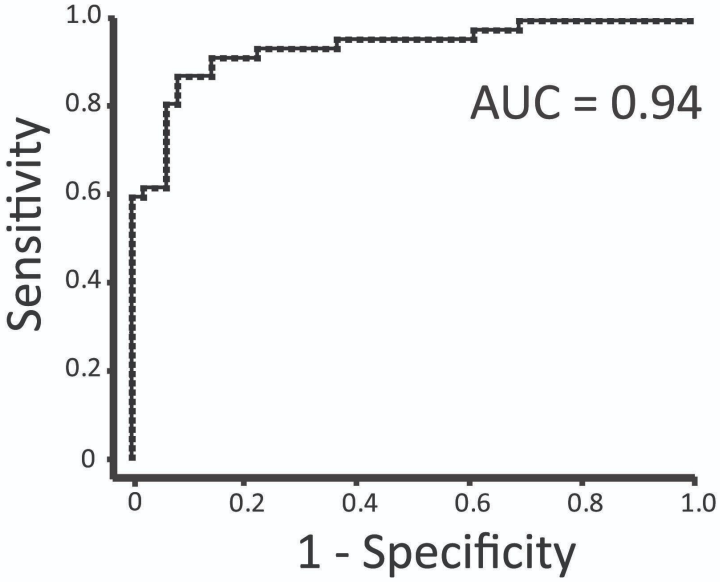
categorical variable prediction

		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

A 2x2 table is a type of confusion matrix

Sensitivity

$$\frac{TP}{TP + FN}$$



Specificity

$$\frac{TN}{TN + FP}$$

Accuracy	What % were predicted correctly?
Sensitivity (Recall)	Of those that *were* positives, what % were predicted to be positive?
Specificity	Of those that were *negatives*, what % were predicted to be negative?
Precision	Of those predicted to be positive, what % were actually positive?
F1 score	One number that balances precision and recall (their harmonic mean)

`scikit-learn`'s `classification_report` shows precision, recall and F1 for every class (you'll read one in A6)

https://forms.gle/AeRuy7WrP6JZWR5t5

Q2: Reading a confusion matrix

A spam filter checks 100 emails: TP = 8, FP = 2, FN = 12, TN = 78. What is its sensitivity (recall)?

A) 40%

B) 80%

C) 86%

D) 97.5%



Classical Machine Learning

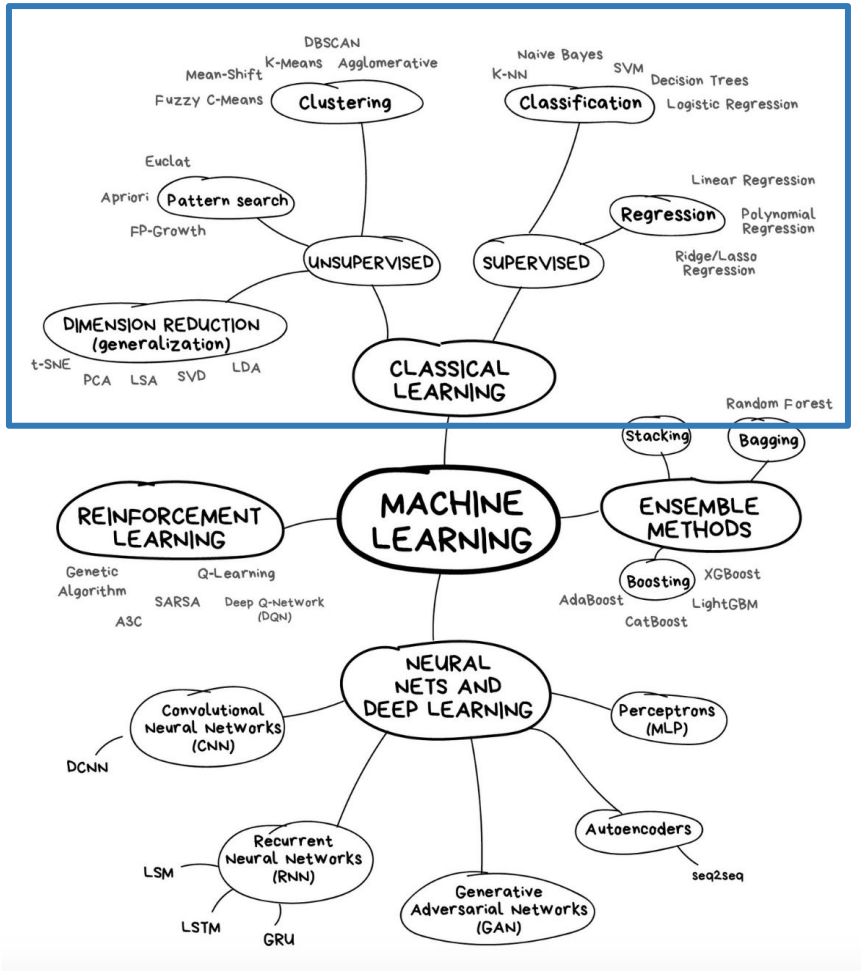


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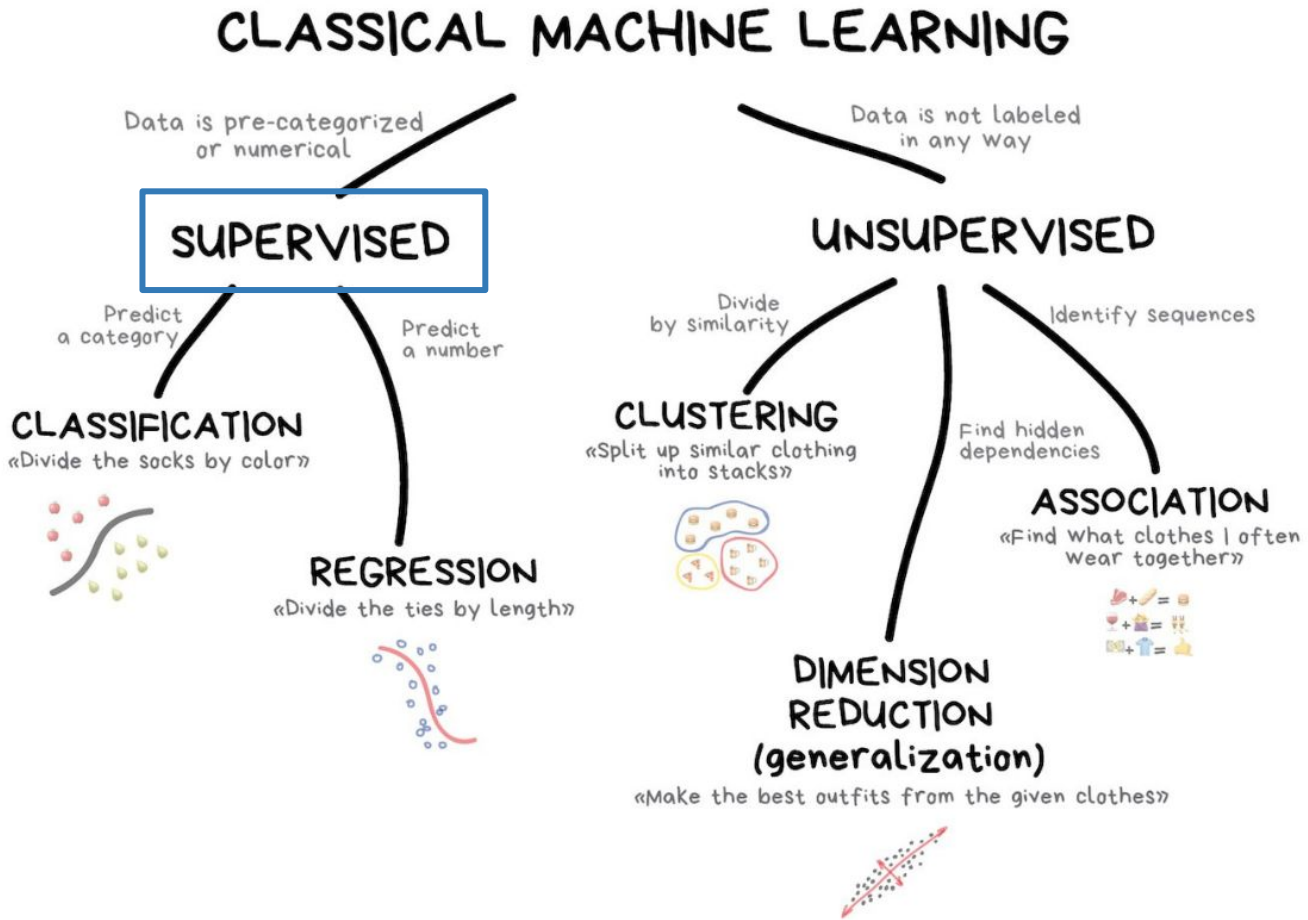


Image source: https://vas3k.com/blog/machine_learning/

Regression

- Predicting a **continuous** outcome by finding a relationship between inputs and outputs
- Makes predictions on new data based on the learned patterns from the training set
- Used for:
  - Stock price forecasts
  - Demand and sales volume analysis
  - Number-time correlations

- Popular regression models include:
  - Linear regression
  - Polynomial regression

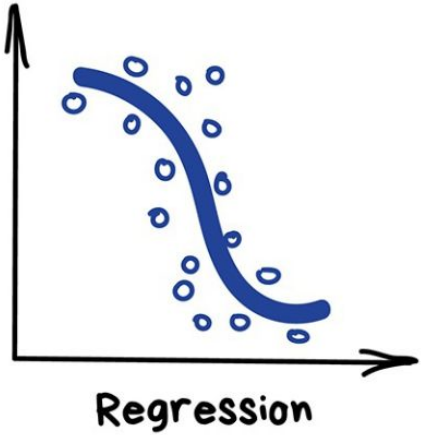
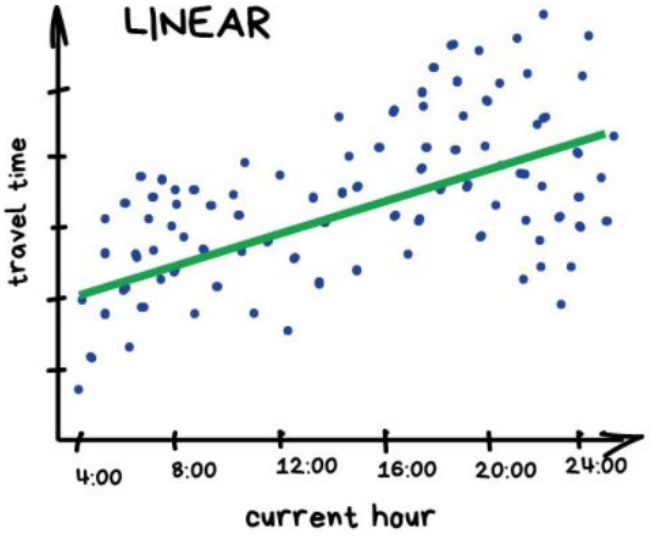
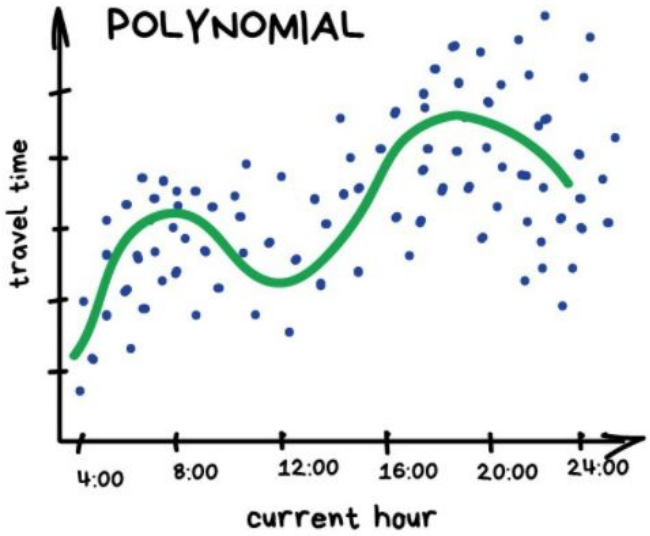


Image source: https://vas3k.com/blog/machine_learning/

Adapted from Sid Joshi's (former COGS 108 TA) slides

PREDICT TRAFFIC JAMS





REGRESSION

Image source: https://vas3k.com/blog/machine_learning/

Classification

- Predicting a **categorical** outcome by finding boundaries between classes of data points
- Makes predictions on new data based on wh ndary it falls
- Used for:
  - Spam filtering
  - Fraud detection

- Popular classification models include:
  - Support Vector Machine
  - Decision Tree
  - K-Nearest Neighbors

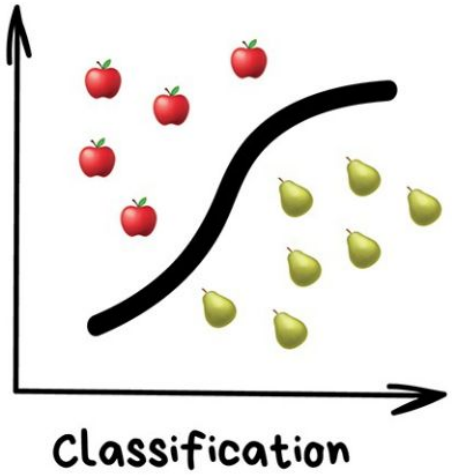
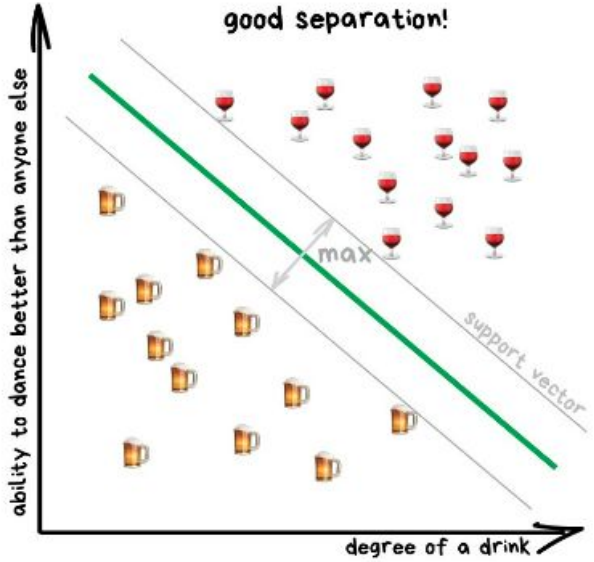


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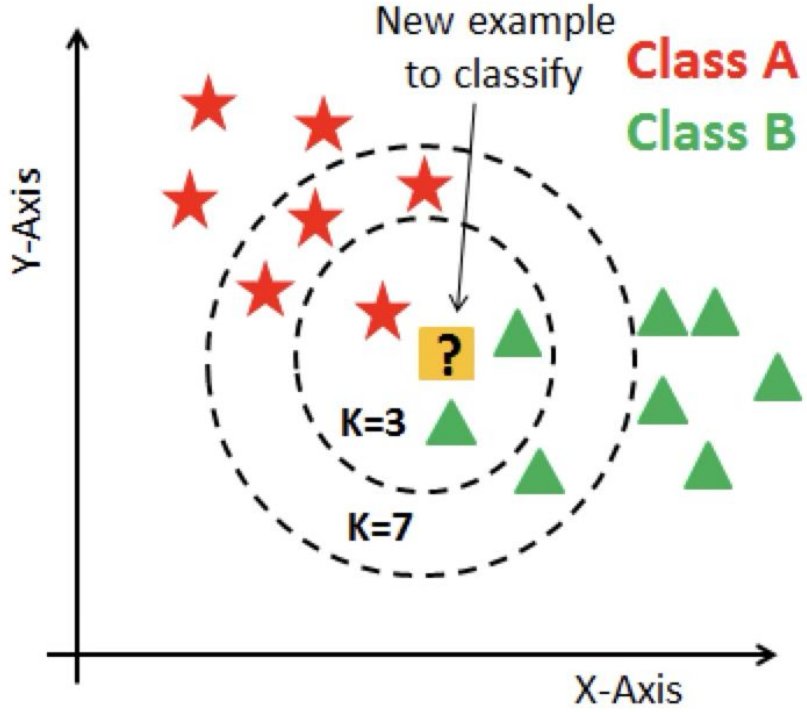
SEPARATE TYPES OF ALCOHOL



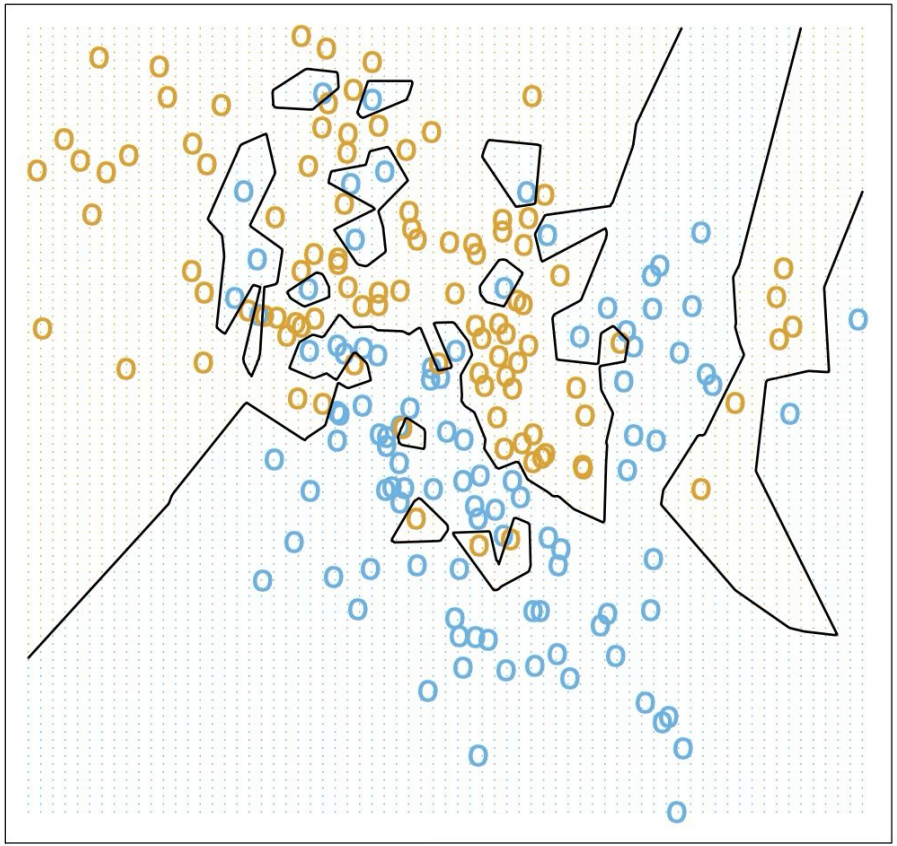
SUPPORT VECTOR MACHINE

Image source: https://vas3k.com/blog/machine_learning/

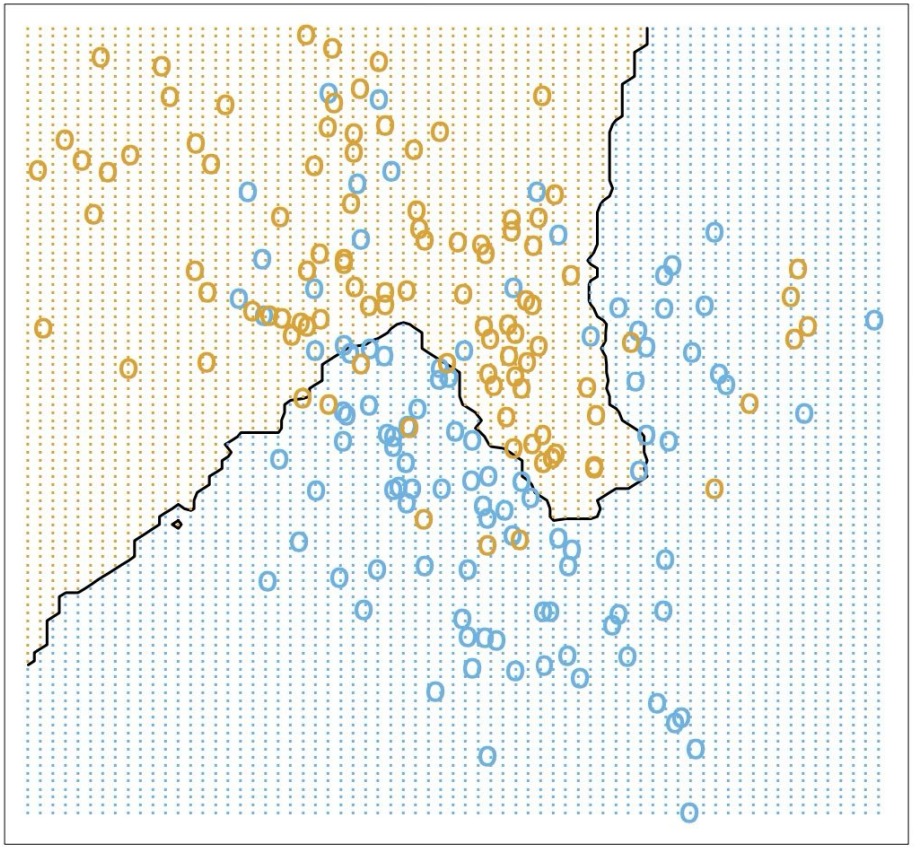
K-nearest neighbors



1–Nearest Neighbor Classifier



15-Nearest Neighbor Classifier



https://forms.gle/AeRuy7WrP6JZWR5t5

Q3: Training vs. test accuracy

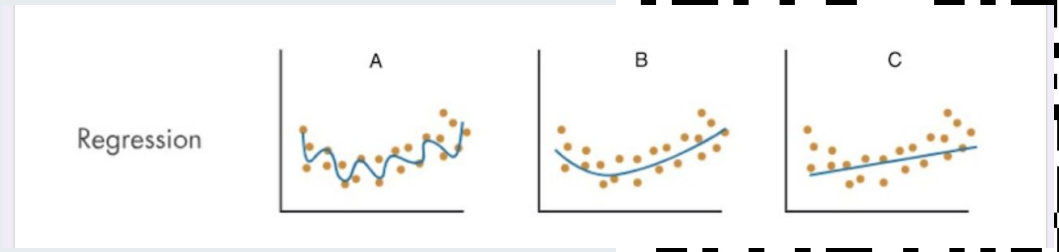
A KNN classifier with k = 1 gets 100% accuracy on the training data and 70% on the test data. This is best described as:

A) Underfitting

B) Overfitting

C) Data leakage

D) A great model



Training accuracy is too optimistic

- With k = 1, every training point is its own nearest neighbor, so training accuracy is (nearly) 100%
- What matters is accuracy on data the model hasn’t seen: validation or test data
- A big gap between training and test performance is a sign of overfitting
- In A6 (2c and 2d) you’ll predict on both and compare

KNN uses distances, so units matter

- Features with large values dominate the distance: elevation (thousands of meters) swamps slope (tens of degrees)
- Scaling (e.g., StandardScaler) puts every feature on the same footing, like the wine example in Lecture 11
- Scaling isn’t automatically better: it also gives weak features as much say as strong ones
- In A6 Part 3 you’ll check whether it helps

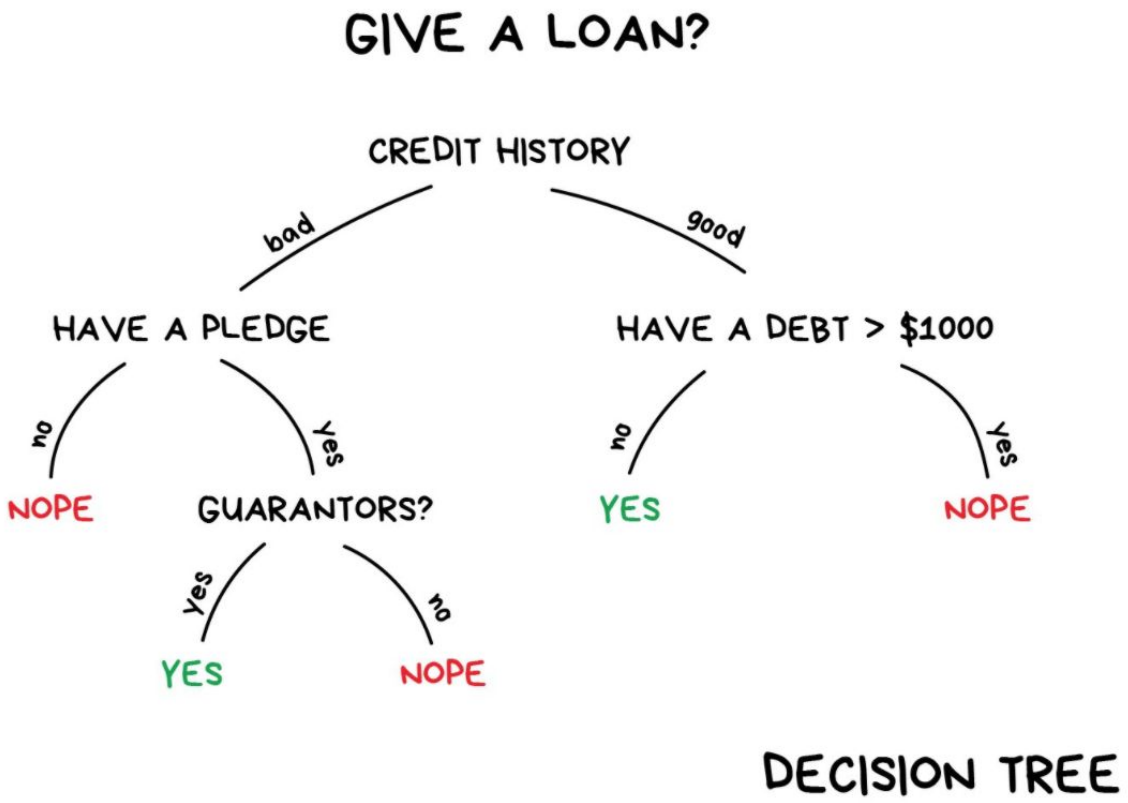
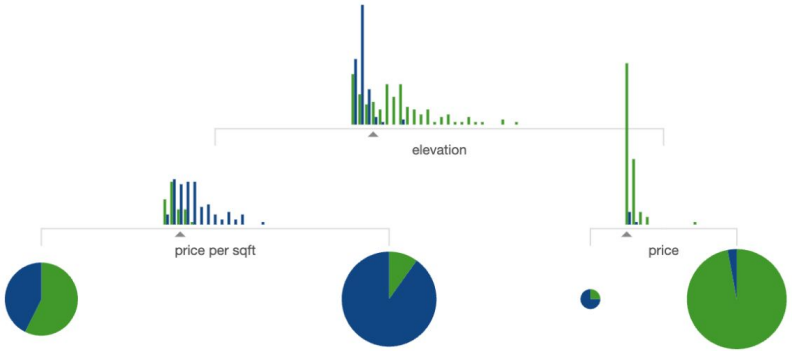


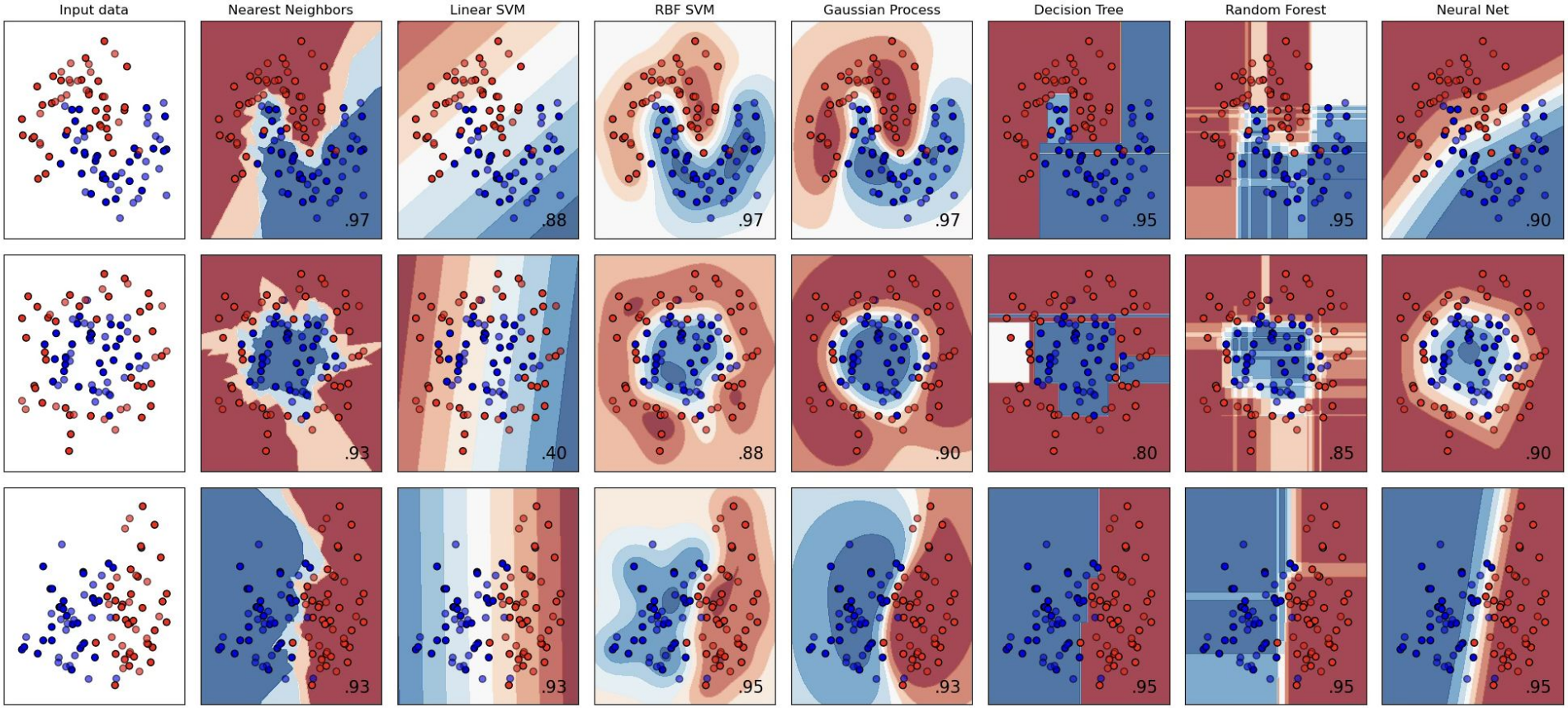
Image source: https://vas3k.com/blog/machine_learning/

Growing a tree

Additional forks will add new information that can increase a tree's **prediction accuracy**.



http://www.r2d3.us/visual-intro-to-machine-learning-part-1/



https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html

Unsupervised Machine Learning

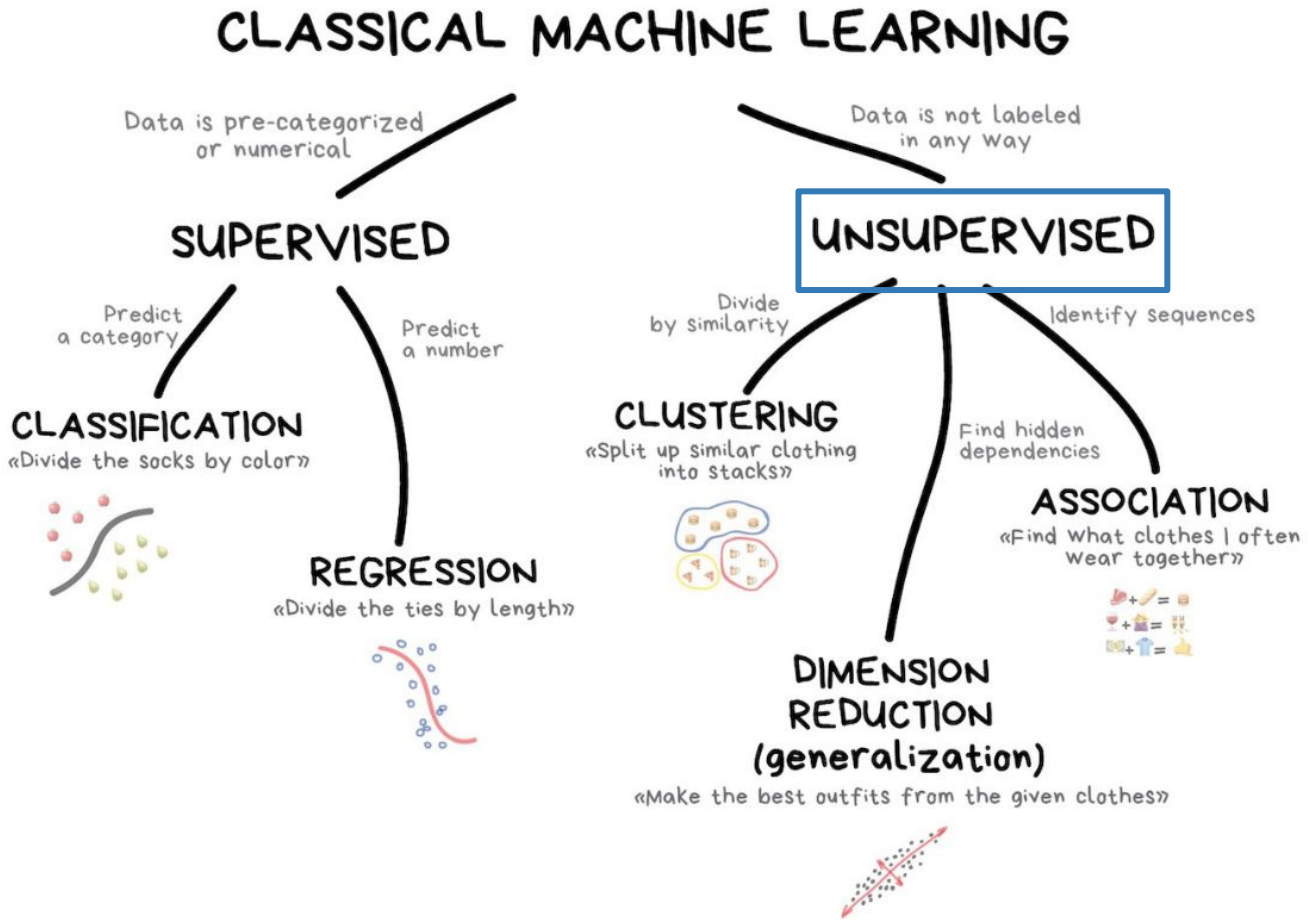


Image source: https://vas3k.com/blog/machine_learning/

Clustering

- Groups data points into clusters based on how **similar** they are to one another
- Similarity is usually defined by spatial relationships
- Used for:
  - Analyze and label new data
  - Anomaly detection
  - Merging points on a map

- Popular clustering models include:
  - K-means clustering
  - Gaussian Mixture
  - DBSCAN

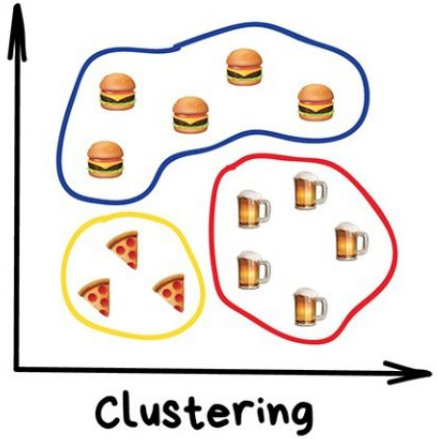


Image source: https://vas3k.com/blog/machine_learning/

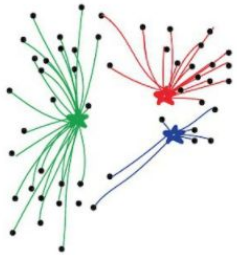
Adapted from Sid Joshi's (former COGS 108 TA) slides

PUT KEBAB KIOSKS IN THE OPTIMAL WAY

(also illustrating the K-means method)



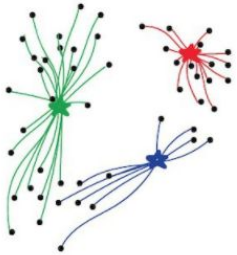
1. Put kebab kiosks in random places in city



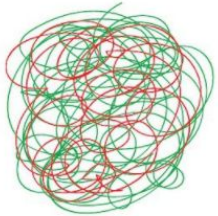
2. Watch how buyers choose the nearest one



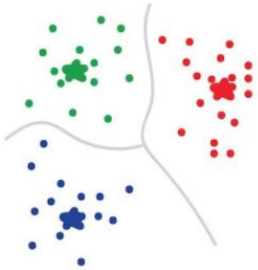
3. Move kiosks closer to the centers of their popularity



4. Watch and move again



5. Repeat a million times



6. Done!
You're god of kebabs!

K-Means

Image source: https://vas3k.com/blog/machine_learning/

Dimensionality Reduction

Covered in depth in Lecture 11

- Reduce high-dimensional data down to fewer dimensions
- Transforms data in a manner that preserves important, explanatory information
- Used for:
  - Fake image analysis
  - Data compression/feature extraction
  - Genomics and bioinformatics

- Popular dimensionality reduction models include:
  - Principal component analysis (PCA)
  - t-SNE
  - UMAP

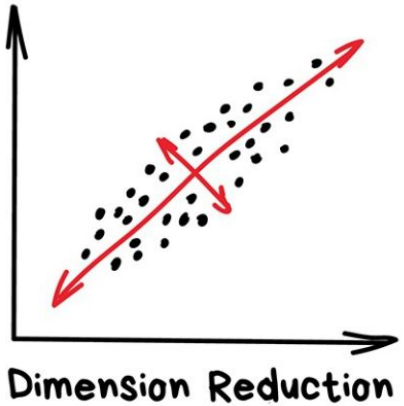


Image source: https://vas3k.com/blog/machine_learning/

Deep Learning and Neural Networks

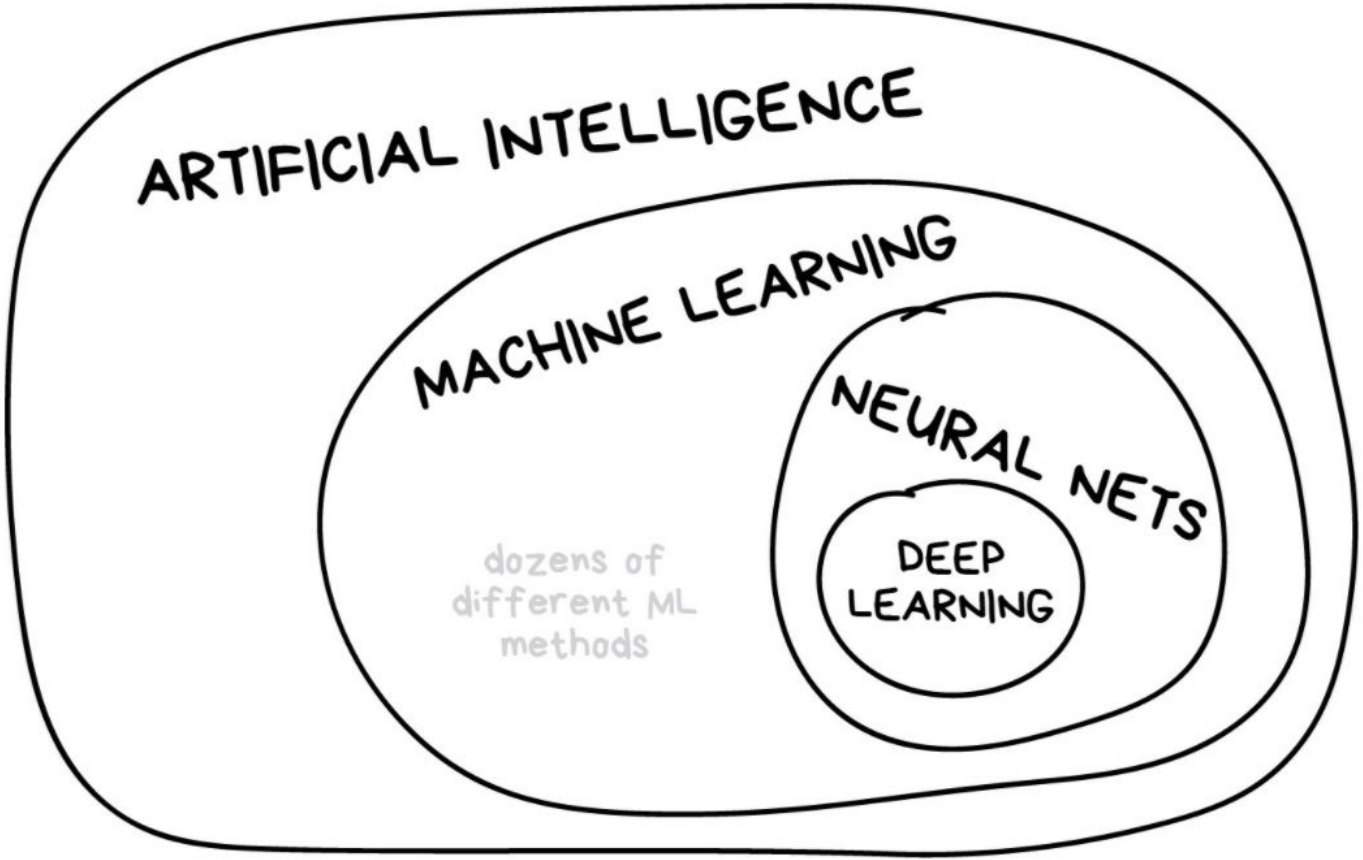


Image source: https://vas3k.com/blog/machine_learning/

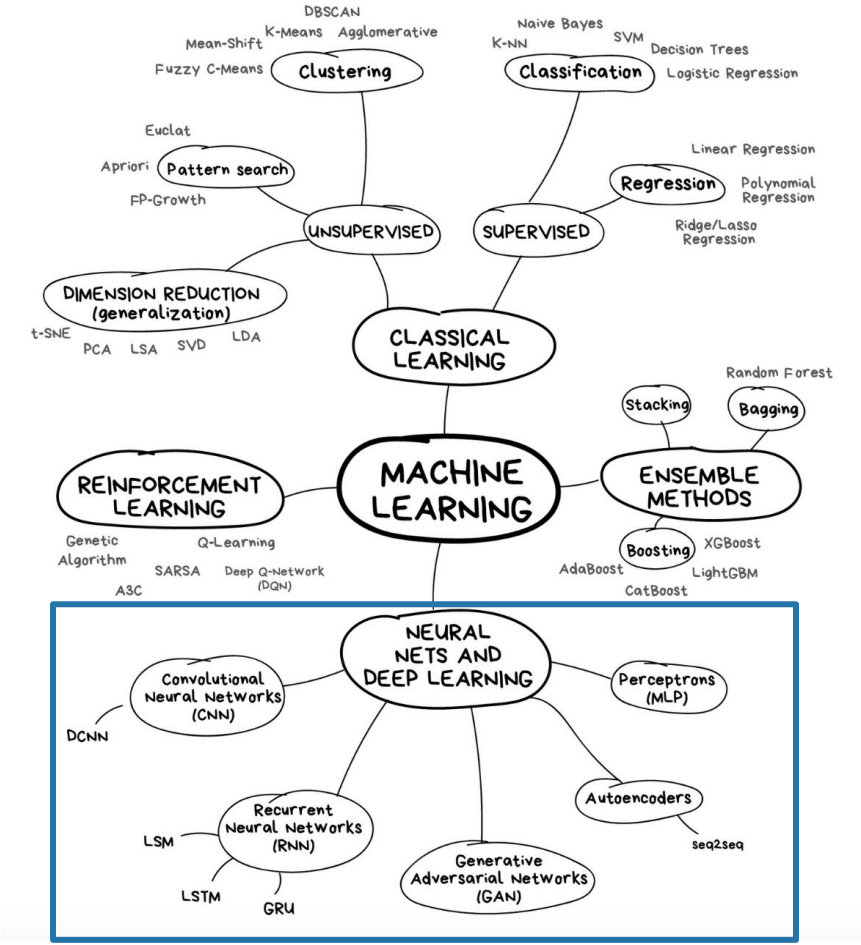
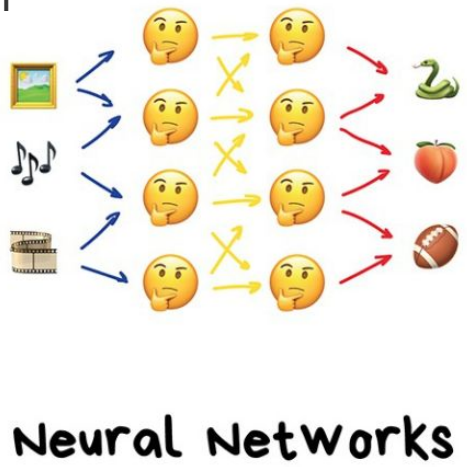


Image source: https://vas3k.com/blog/machine_learning/

Neural Networks

- An arrangement of interconnected nodes inspired by the human brain that process information to learn and make predictions
- **Can** learn any of these tasks, but aren't always better
- Used for:
  - Image segmentation
  - Speech recognition
  - Image/text/video generation
- Popular neural network architectures include:
  - Multi-layer Perceptron
  - Convolutional Neural Networks
  - ResNet, Transformers (used in LLMs)



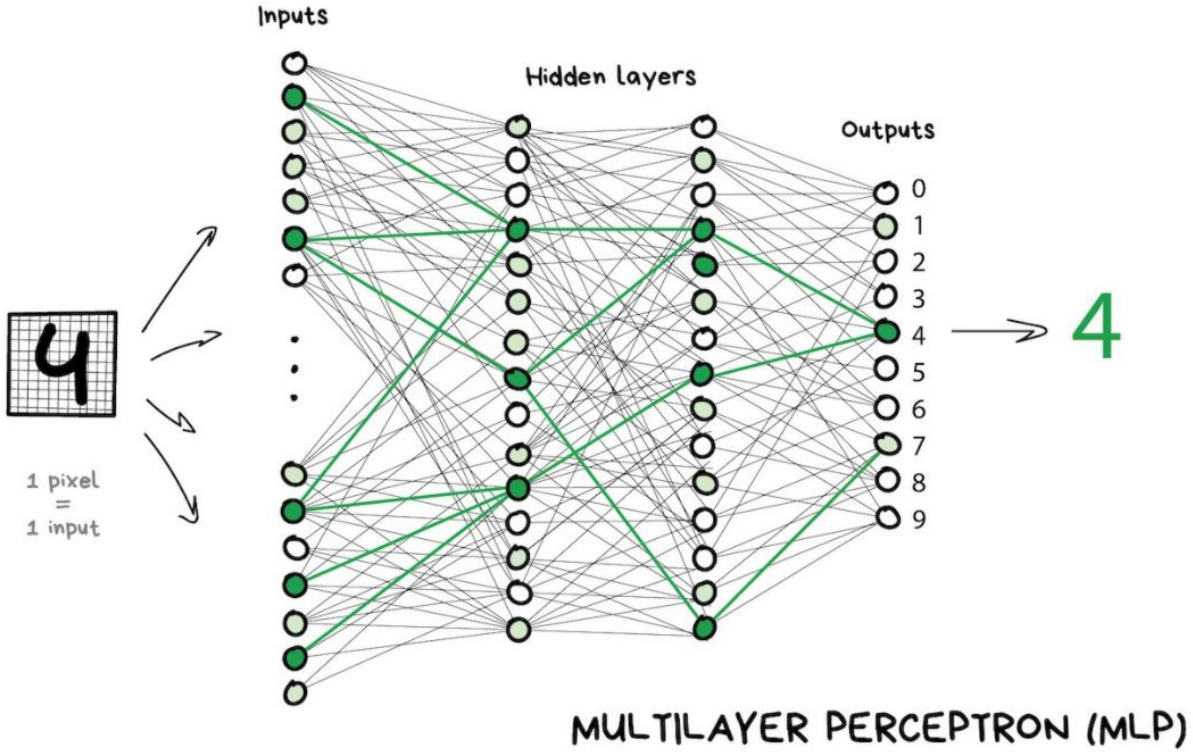
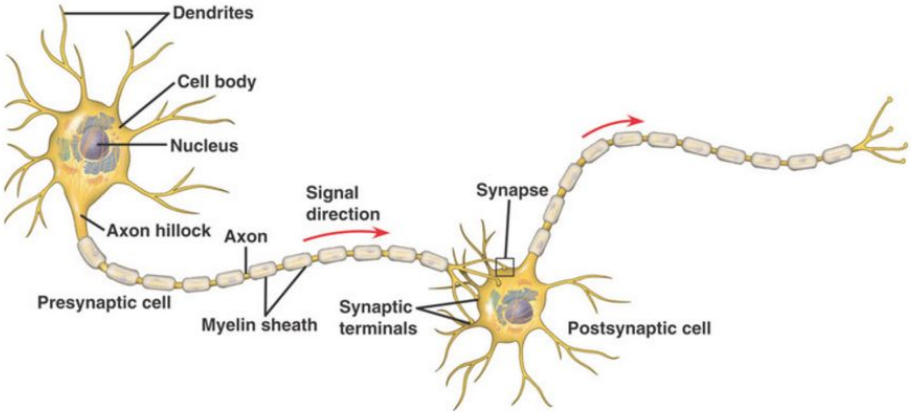


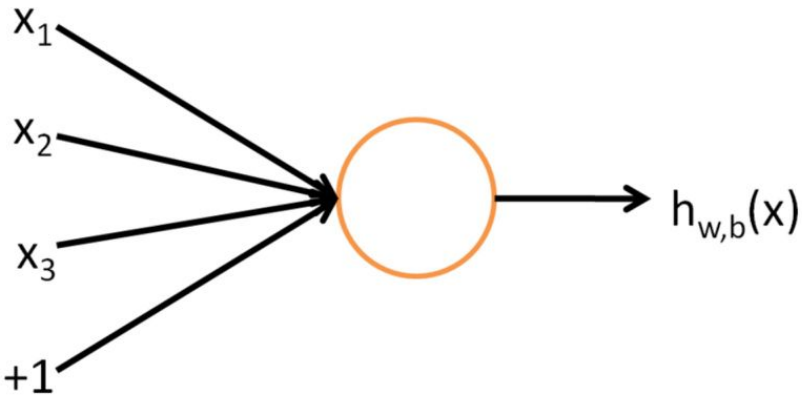
Image source: https://vas3k.com/blog/machine_learning/

WHAT IS A NEURON?



- Receives signal on synapse
- When trigger sends signal on axon

MATHEMATICAL NEURON



- Mathematical abstraction, inspired by biological neuron
- Either on or off based on sum of input

These weights tell the neuron to respond more to one input and less to another. Weights are adjusted when training — that's how the network learns. Basically, that's all there is to it.

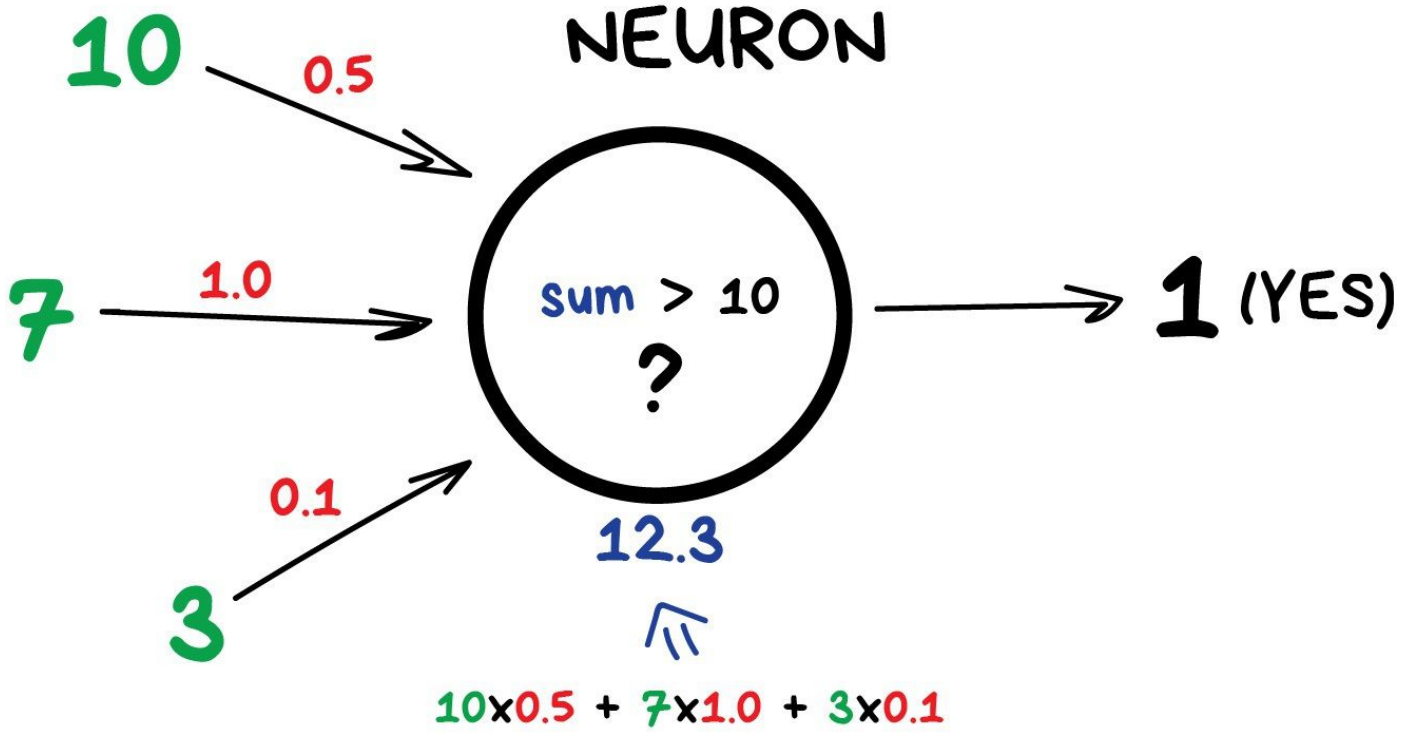


Image source: https://vas3k.com/blog/machine_learning/

The challenges of image processing

- Extracting features from images was (historically) **hard**
  - Historically, humans would manually label images with particular features → handcrafting features

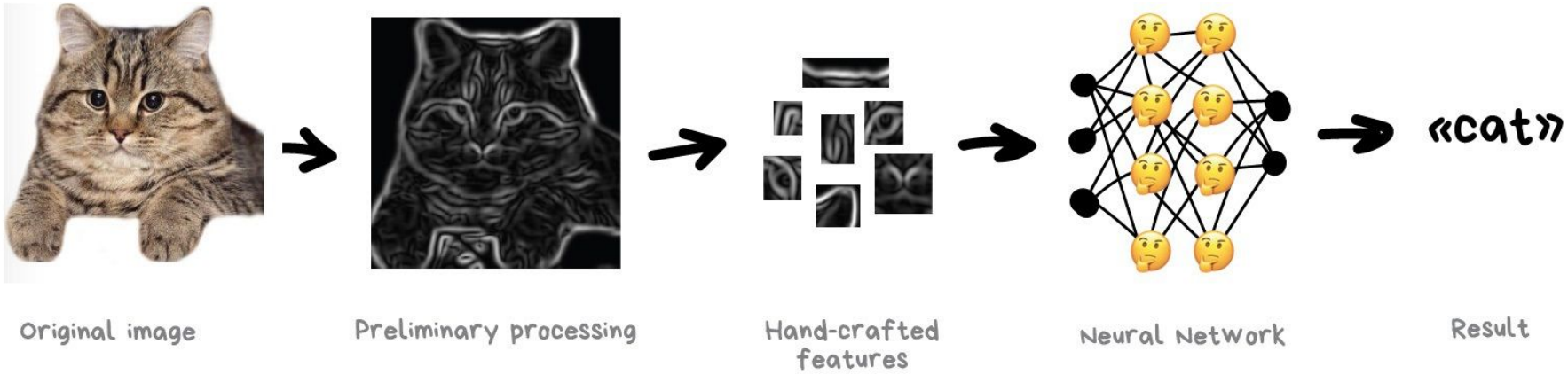
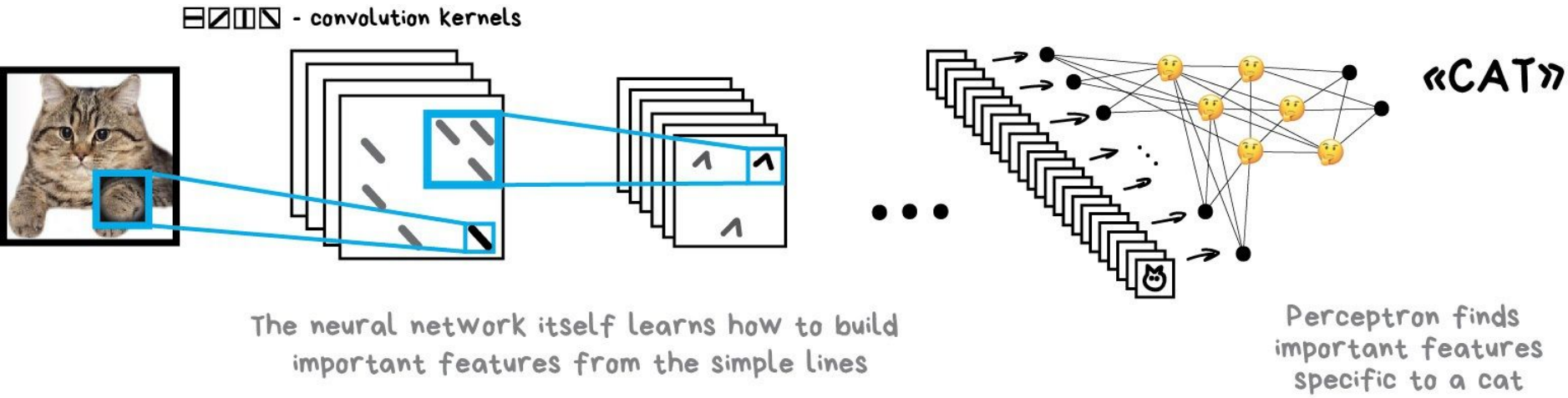


Image source: https://vas3k.com/blog/machine_learning/

CNNs to the rescue!



CONVOLUTIONAL NEURAL NETWORK (CNN)

Image source: https://vas3k.com/blog/machine_learning/

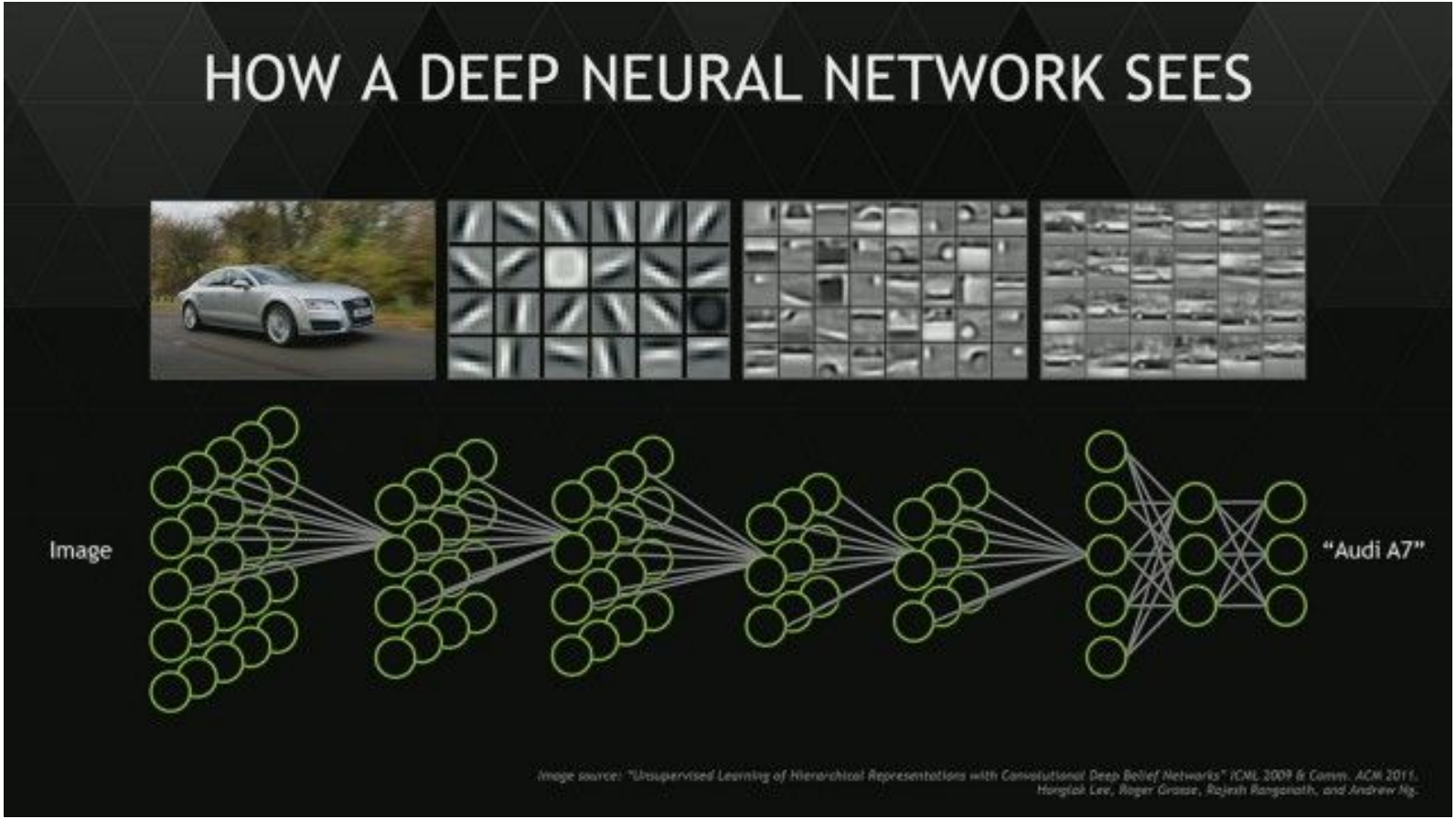


Image Source: https://towardsdatascience.com/understanding-residual-networks-9add4b664b03

Large language models

- Neural networks built on the transformer architecture (2017), now used for text, images, audio and code
- Trained to predict the next token across a huge amount of text: the labels come from the text itself
- ChatGPT, Claude and Gemini start this way, then are fine-tuned with human feedback
- Today’s ideas still apply: the training data shapes what the model learns, and it can be confidently wrong on inputs unlike its training data

Machine Learning Ethics

What to do about ethical concerns and biased algorithms...

1. Anticipate and plan for potential biases before model generation. Check for bias after.
2. Have diverse teams.
3. Use machine learning to improve lives rather than for punitive purposes.
4. Revisit your models. Update your algorithms.
5. You are responsible for the models you put out into the world, unintended consequences and all.

Think about whether the models you’re building should even be built.

https://forms.gle/AeRuy7WrP6JZWR5t5

Q4: Is it fair?

Consider an early-warning model that flags students likely to fail was “81% accurate.” (discussed week 2) **Which question best checks whether it’s fair?**

A) Can we get it to 90%?

B) What are the false positive and false negative rates for each group of students?

C) Which algorithm did they use?

D) How many features does it use?

“81% accurate”: who are the other 19%?

- Week 2: Canvas activity predicts who will fail by week 3
- A false positive labels a student “at-risk” who wasn’t, and the label can change how they’re treated
- A false negative misses a student who needed help
- Login patterns track jobs, commutes and internet access, so error rates may differ by group
- Check the confusion matrix for each group, not just overall accuracy
- *Also*: should it be built? If yes, why? If no, why not?

Predictive algorithms should (*at a minimum*) be FAccT

Fair: lacking biases which create unfair and discriminatory outcomes

- For whom does this algorithm fail?
- Steps to take:
  1. Verify data about individual is correct
  2. Carry out “sensitivity test”

Accountable/Accurate: answerable to the people subject to them

- Correct data used? Is there a mechanism for appeal?

Transparent: open about how and why particular decisions were made

- Think *carefully* about what transparency is (Handing over source code likely isn’t the answer)

Sources: https://www.datacamp.com/community/podcast/weapons-math-destruction & https://dl.acm.org/doi/fullHtml/10.1145/3290607.3310433

A Mulching Proposal

Analysing and Improving an Algorithmic System for Turning the Elderly into High-Nutrient Slurry

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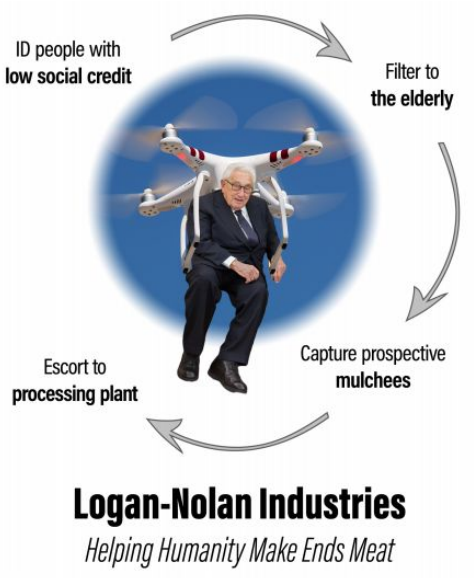


Figure 1: A publicity image for the project, produced by Logan-Nolan Industries

https://dl.acm.org/doi/fullHtml/10.1145/3290607.3310433 (*CHI* 2019)

A Mulching Proposal

FAIR - equally considers all elderly individuals

ACCURATE - pre- has mechanism for appeal; post - compensation

TRANSPARENT - website with all features; testable

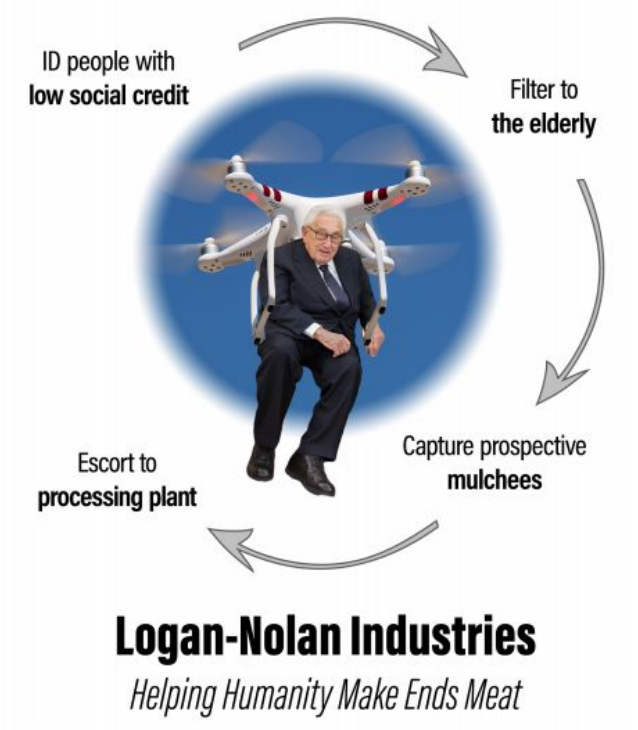


Figure 1: A publicity image for the project, produced by Logan-Nolan Industries

https://forms.gle/AeRuy7WrP6JZWR5t5

Q5: Prediction Thoughts

We should start using this algorithm to mulch up the elderly.

A) Strongly Agree

B) Somewhat Agree

C) Neutral

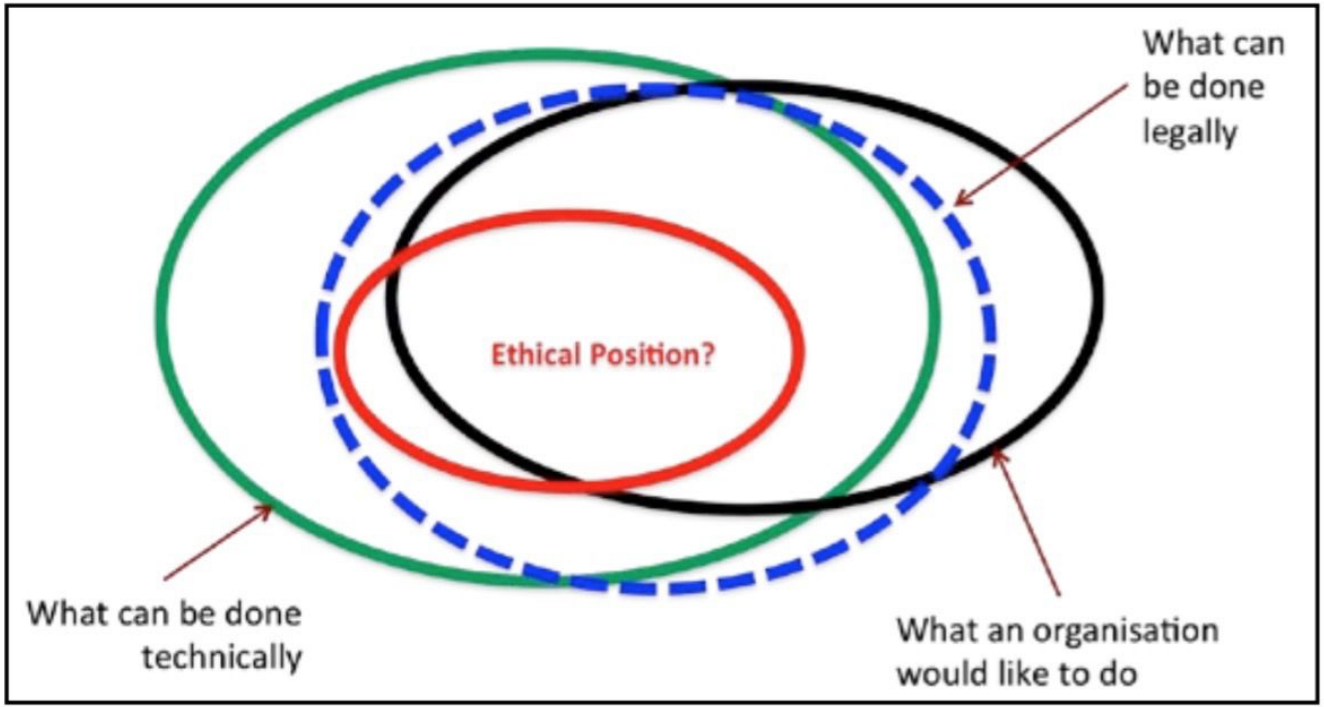
D) Somewhat Disagree

E) Strongly Disagree



Checklists are helpful, but they’re not an excuse for thoughtlessness.

Big Data Ethics



Source: IBM

Beyond COGS 108

- **Supervised Learning - COGS 118A**
  - Labelled data, predict, classify, or fit a model
- **Unsupervised Learning - COGS 118B**
  - No labels, no feedback
  - Find hidden structures in a model
- **Reinforcement Learning - COGS 182**
  - No labels or rules; exploring an environment
- **Neural Networks - COGS 181**
  - Deep Learning and CNN architectures
- **Genetic Algorithms - COGS 186**
  - Mimicking evolution in an agent through trial-and-error
  - Also: COGS 185 (Advanced ML Methods) and COGS 188 (AI Algorithms)